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STOCK PRICE ANALYSIS AND PREDICTION

MSc. Data Analytics

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Abstract

In this study, the analysis and predictions of Apple Inc. stock prices will be done using machine learning approaches and data visualization technique. Predicting stocks is quite complicated since the financial markets are extremely dynamic and unpredictable. The major goal of this research is to analyse the ability of different prediction models based on machine learning and improve interpretation of results by applying visualization.

The historical stock price data have been gathered from the website of Yahoo Finance using Python. The data includes stock indicators like opening price, closing price, high price, low price, and trading volume. The data was cleaned, prepared and exploratory data analysis was done.

Two predictive models have been used for stock prices prediction. The first one is Linear Regression which is the base model to compare results with other models, and another is Long Short-Term Memory (LSTM) model. The models' performance was analysed by using the values of root mean square error (RMSE) and mean absolute error (MAE).

According to the result, the linear regression model was able to determine any overall trend, though prediction accuracy could not be achieved due to the fluctuations in the market. However, the LSTM performed better because it can analyse complex dependencies.

Moreover, Power BI dashboards were created to graphically visualize trends in stock performance and prediction accuracy. In conclusion, machine learning methods have been seen as efficient means of analysing and predicting stock movement. However, there may be issues in using historical data only, leading to lower prediction accuracy.

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1. Introduction

1.1 Introduction

The stock market forms an integral component of the financial system since it enables firms to generate funds while providing investment prospects for investors. Stock price prediction remains a key area of interest to both researchers and investors who target on making informed investment choices and avoiding potential losses. On the other hand, stock market prediction remains challenging because stock prices may be influenced by various elements, including market volatility, investor sentiments, economic environment, politics, among others (Fama, 1970).

Due to increased access to financial data and technological advancement, huge applications of machine learning and deep learning models have been developed to support stock market forecasting (Domingos, 2012) .

The current project deals with the analysis and prediction of stock prices of Apple Stock. as it is characterized by its dominance on the market and availability of its historical data on Yahoo Finance. The study considers two prediction methods: Linear Regression as a Machine Learning method and Long Short-Term Memory (LSTM) as a Deep Learning method. For that, data processing included data cleaning and feature engineering with the use of such techniques as moving average and calculating return (Elisseeff, 2003).

Moreover, interactive dashboards with visualizations of stocks' tendencies, market behaviour, technical indicators, volumes of trading and prediction results were created by means of Microsoft Power BI Desktop. This study integrates programming, machine learning, deep learning and visualization.

1.2 Business Understanding

The stock market is an important part of the world financial market and serves the purpose of investments and growth. People are always looking at different factors that influence stocks to increase their efficiency and minimize risks. However, stock prices are very volatile and depend on different economic parameters.

This research is designed to investigate the behaviour of the stock market and forecast future values for stock prices based on past records. This paper considers stocks of **Apple Inc. (AAPL)**. and uses two methods to build predictive models: linear regression and LSTM approach. In addition, dashboards are developed in Microsoft Power BI Desktop to show changes and forecasts for stock prices (Kandel, 2023).

1.3 Research Objectives

To accomplish the objective of this research, the following research objectives have been created:

- Collecting stock prices data from a reputable data source.

- Cleaning the collected data for analysis purposes.
- Conducting exploratory data analysis on the collected dataset
- Modelling stock price prediction using Linear Regression and LSTM.
- Testing the effectiveness of the models through appropriate evaluation measures.
- Implementing a Power BI dashboard for visualizing the results.
- Analysing and critically reviewing the results obtained.

1.4 Research Questions

The following research questions will be answered by this study:

a) How well do machine learning algorithms perform in predicting the price of stocks?
b) Is the performance of the LSTM algorithm better than that of Linear Regression?
c) What conclusions can be drawn from the exploration of the stock dataset?
d) What are the benefits of using visualized data in analysing stock price trends?

1.5 Importance of the Study

The importance of this study is diverse since it combines the application of financial analysis, machine learning algorithms, and data visualization tools under one umbrella framework for predicting stocks price in the financial markets.

Academic Significance

The study will significantly contribute to the emerging literature on the application of machine learning and deep learning algorithms in the field of finance (J. B. Heaton, 2016). In particular, the project examines the applicability and reliability of different models and their respective advantages and disadvantages for the predicting stock prices. The research will help in gaining a better understanding of time-series analysis and pre-processing of financial data that may assist future research in analysing data.

Practical Significance

The findings of this study can be helpful to investors and analysts who wish to understand the behaviour and performance of stocks in the market. Moreover, the use of historical data for the purpose of predicting future trends will prove valuable in making good investment decisions.

Technical Importance

Combining machine learning algorithms with Microsoft Power BI allows for presenting the practical usage of modern data analytic technologies in financial prediction tasks. The implementation of the project proves that the combination of predictive models created using Python and visualization methods can become a basis for an effective analysis system.

1.6 Scope of the Study

In this study, the priority will be on the analysis and prediction of the stock price of Apple Inc. from historical stock market data. This research work requires analysing the time series of stock price data historically to make trend predictions, patterns, and behaviour of the stock market within a certain time period. The application of machine learning and deep learning methods like linear regression and LSTM to predict future stock price based on previous observations is considered in this study. Also, the visualization of the results of the analysis and predictions is carried out in the form of dashboards using Microsoft Power BI Desktop (Kandel, 2023). Nevertheless, the scope of this work is limited to historical stock market data only without considering other factors affecting the stock price.

1.7 Limitations of the Study

Even though the project manages to analyse and predict stock market performance, there are a few limitations of project. The first constraint is that this study depends mostly on historical data from the stock market to predict future prices. However, the accuracy of such predictions may be imperfect since stock markets are very unpredictable. There is also the problem of considering the influencing factors that impact stock prices, such as political and economic events and the company's performance, which is hard to be considered for the researchers. The next constraint relates to computational capacity. This is because models like LSTM require high computing capacity to operate efficiently. Lastly, the study only takes into consideration the prices of Apple Company’s stocks, which restricts generalization.

1.8 Project Tools and Technologies

The project uses the following tools and technologies:

- Python as the main programming language
- Libraries Pandas, NumPy, Scikit-learn, TensorFlow
- Software Power BI Desktop

2. Literature Review

2.1 Introduction

The topic of stock market prediction has been widely studied within the world of financial analytics and machine learning because of its importance in investment decisions and risk management. Number of researchers have utilized various techniques to improve predictive modelling and understand stock markets better. Financial markets are very dynamic and subject to huge economic, political, and psychological influences, making stock prediction complicated.

In this chapter, the literature is critically analysed on stock market prediction, machine learning algorithms, deep learning methods, feature engineering, and data visualization. These sources will be valuable for the present study and help to determine research gaps concerning the use of machine learning and Power BI visualizations in stock price predictions.

2.2 Review of Previous Literatures

2.2.1 A Critical Review of Recurrent Neural Networks for Sequence Learning

Zachary C. Lipton and John Berkowitz (2015) studied the application of Recurrent Neural Networks (RNNs) for sequential and time-series prediction tasks. The authors noted that RNN models were built for sequential datasets because they had the ability to retain information about previous observations in the prediction process. In contrast to standard neural networks, RNN models have internal memory which makes it capable of affecting predictions by using the previous information, which makes them very effective in prediction.

Moreover, according to the literature review conducted by the authors, RNN models are significantly effective at recognizing the dependence among financial data, also identifying trends on the market. It was mentioned that modern advanced RNNs, such as LSTMs, work much better in terms of accuracy and efficiency as compared to ordinary RNNs because they are not affected by the problem of vanishing gradients.

2.2.2 Forecasting Economic and Financial Time Series: ARIMA vs LSTM

Sima Siami Namin and Akbar Siami Namin (2018) done a comparative study between ARIMA and LSTM models for time-series forecasting and financial prediction. To investigate the performance of both algorithms in terms of prediction, the scientists employed sequential datasets and demonstrated that LSTM provided higher levels of accuracy rates than ARIMA. One should admit that according to the source, ARIMA can be applied exclusively to linear and stationary data since financial markets possess dynamic characteristics.

Moreover, one should point out that in contrast to ARIMA that requires the application of strict statistical assumptions, LSTM models utilize specific memory cells and gates to store information about previous occurrences and use it during analysis of new datasets, which makes them more advantageous for forecasting. Hence, one may conclude that neural networks and other approaches to deep learning are superior to ARIMA in terms of predicting stock prices.

2.2.3 Long Short-Term Memory

Long Short-Term Memory model was developed by Sepp Hochreiter and Jurgen Schmidhuber as an improvement on the classical Recurrent Neural Networks model in 1997. Their study emphasized the necessity to solve the issue with vanishing gradients to enhance the capability of the classic RNNs to learn long-term dependencies from sequential data.

The authors came up with the concept of LSTM by integrating the memory cells along with the input, forget and output gates into their model. It provided the ability to store the relevant data for LSTM. LSTM proved itself as a more effective model at learning long-term dependencies compared to RNN. As a result, LSTM was successfully applied to stock market forecasts and financial modelling.

2.2.4 An Introduction to Variable and Feature Selection (Feature Engineering and Financial Prediction)

Isabelle Guyon and Andr´e Elisseeff (2003) explored the importance of feature engineering and feature selection in machine learning. The authors indicated that the choice of input features enhances the quality of the model, its accuracy of forecasts, and computational speed.

This study highlighted that feature engineering plays a crucial role in financial prediction due to the high level of noise and non-linear dependencies within financial market datasets. Correct feature engineering facilitates the detection of trends by machine learning models. Feature engineering methods utilized in the current research projects are highly recommended by this study.

2.2.5 Efficient Market Hypothesis

Eugene F. Fama presented the Efficient Market Hypothesis (EMH) in 1970. The Efficient Market Hypothesis says that stock prices completely capture all the information available in the market. This theory holds that consistent earning excesses above the market average cannot be attained since stock prices instantly change when information becomes available.

The research truly affected contemporary finance theory and analysis of stock market performance. However, opponents of the theory pointed out that stock markets tend to act irrationally, emotionally, and inefficiently, providing sufficient room for predictive analytics and machine learning techniques.

The literature is significant since it addresses the conceptual issue involved in predicting stock prices, explaining the complexities of stock price prediction.

2.2.6 Data Visualization and its Impact in Decision Making

The research done by Sharad Sharma and Archita Kandel (2023). explains the significance of data visualization in the decision-making process and the importance of converting complicated data into visual forms such as graphs, reports, and dashboards. The study focuses on the increased usage of business intelligence tools like Microsoft Power BI Desktop for analysis. Visualization is said to be crucial in financial analytics because it helps to analyse the trend of stocks, volatility, and prediction results.

2.2.7 What is Power BI

Microsoft launched Microsoft Power BI Desktop as a tool for business intelligence and visualizations which supports in transforming raw data into interactive dashboards and reporting. A few researchers working on business intelligence systems argued that Power BI helps in improving analytical reporting and decision-making through interactive visualization.

Some key benefits of using Power BI were discussed in the literature review:

- Interactive dashboards
- Data integration
- Visualization in real time

- Business reporting
- Analytical communication

2.2.8 Deep Learning

Yann LeCun, Yoshua Bengio and Geoffrey Hinton in 2013 discussed the growing role of deep learning in sequential data analysis and forecasting tasks. According to the researchers, deep learning algorithms are very powerful in discovering the underlying structures or relationships in data through self-learning of non-linear relationships without the need for any human intervention in feature extraction.

The conclusion made by the research was that deep learning methods were superior to statistical techniques in dealing with complex financial data analysis.

2.2.9 A few useful things to Know about machine Learning

Pedro Domingos (2012) discussed the practical applications of machine learning in business analytics and financial forecasting. In the study, it was indicated that machine learning algorithms have become common in predicting stock market price because they can analyse big data and discover any trends in the stock market more efficiently than traditional methods of analysis. This is possible since traditional analytical methods face difficulty when analysing complex and large financial dataset.

Additionally, the need for efficient data preprocessing and feature engineering was noted in the study. It was mentioned in the paper that the quality of the input data influences greatly the efficiency of machine learning technologies in finance, hence, the need for effective data cleaning and preprocessing. Lastly, it was noted in the study that machine learning has revolutionized financial forecasting in today's world due to the possibility to make faster and informed investments.

2.2.10 Deep Learning in Finance

J. B. Heaton, N. G. Polson and J. H. Witte (2016) studied the use of deep neural networks for financial market prediction and investment analysis. The researchers have stated that deep learning models have an ability to capture complex financial relationships, leading to higher prediction accuracy as compared to conventional forecasting methods. It was pointed out in the study that financial markets tend to produce large amounts of both structured and unstructured data, and this makes the application of deep learning models highly suitable because of their ability to automatically learn patterns and features from past data.

The researchers further stated that deep neural networks can deal with large datasets from financial markets in a much better way as compared to traditional machine learning and statistical models. It has been observed that deep learning models performed very well when it comes to the detection of trends, volatility, and serial dependencies in financial time series data.

2.2.11 Stock market's price movement prediction with LSTM neural networks

According to David M. Q. Nelson, Adriano C. M. Pereira, and Renato A. de Oliveira (2017), there was an analysis carried out on the application of deep learning techniques for stock forecasting purposes, whereby LSTM models were compared to machine learning techniques and random walk models in predicting stock returns. It emerged that LSTM was a superior technique in dealing with the task at hand owing to its ability to model complex relationships and temporal dependencies in financial datasets. Additionally, it is worth mentioning that LSTM is well-suited for sequential data and volatility in the market environment.

2.2.12 A deep learning framework for financial time series using stacked autoencoders and Long-Short Term Memory

Wei Bao, Jun Yue and Yulei Rao (2017) have explored the application of deep learning models to forecast stock prices, comparing their efficiency with classical regression analysis methods. It was stated that LSTM demonstrated higher forecasting accuracy and better recognition of trends compared to Linear Regression. The authors reasoned that Linear Regression model cannot deal with the complexity of data related to the stock market, while LSTM model possesses an ability to store history data and extract long-term patterns.

2.2.13 Machine Learning Techniques and Data for Stock Market Forecasting

A review of Mahinda Mailagaha Kumbure, Christoph Lohrmann, Pasi Luukka and Jari Porras (2022) revealed an extensive review of past literature on application of machine learning techniques for stock market prediction. The research examined various past literature on financial forecast models and concluded that there is increasing usage of machine learning models since traditional models have challenges when applied due to dynamics and complexity of financial data. The study also highlighted that features like historical stock prices, technical indicators, and feature engineering play a significant role in increasing accuracy of predictions.

Another important observation of the study was regarding superior performance of deep learning models like LSTM network since they could capture sequential and non-linear relationships within the financial data. It was concluded that integration of machine learning models with technical indicators and other data processing methods leads to increase in accuracy of forecasts. Literature reviewed above is supportive of implementation of feature engineering, Linear Regression model, LSTM models, and stock trend analysis for predicting stock prices of Apple Inc. in the current project.

2.2.14 Time Series Forecasting with Deep Learning: A Survey

Bryan Lim and Stefan Zohren (2021) studied the deep learning techniques used in financial time-series forecasting and stock market prediction. Lim & Zohren reviewed a few deep learning techniques such as RNN, Long Short-Term Memory model (LSTM), CNN, and transformer among others. According to this study, deep learning techniques were efficient when it comes to forecasting since they could learn from the historical information available.

It was evident from the study that the LSTM techniques had proved to be more efficient due to their ability to retain long term information and the sequential relations between them. The study focused on the significance of feature engineering and stock analysis techniques for forecasting and improving performance. Therefore, it is safe to say that the combination of the deep learning techniques and visualization of stocks has led to improved forecasting performances. It can be deduced that the current study will benefit from this literature review.

2.2.15 Stock Market Prediction via Deep Learning Techniques

According to inan Zou, Haiyao Cao, Lingqiao Liu, Yuhao Lin, Ehsan Abbasnejad, and Javen Qinfeng Shi (2022), deep learning models applied to stock market prediction and financial forecasting were examined. In particular, the authors examined RNN, LSTM, CNN, and hybrid models used to handle stock market data. The results showed that deep learning outperforms the existing forecasting tools since they can learn complex relationships between variables of financial data.

It was mentioned that past price levels, technical indicators, and preprocessing enhance prediction accuracy. Among models described by the author, LSTM model was one of the best-performing due to its ability to retain sequence information and learn long-term dependencies.

Thus, it can be concluded that deep learning models significantly improve stock forecasting quality and enhance financial analysis. This literature supports the use of LSTM models in current research.

2.2.16 Stock Price Prediction Using Machine Learning and LSTM-Based Deep Learning Models

Sidra Mehtab, Jaydip Sen and Abhishek Dutta (2020) conducted research on the use of hybrid deep learning approach for prediction in stock markets. In their discussion, the authors stated that using multiple forecasting approaches is effective for increasing accuracy because various models have unique strengths and weaknesses. The study aimed at analysing the application of advanced neural networks for stock price prediction.

According to the findings, hybrid models were more efficient compared to separate models since they could learn both linear and non-linear relationships in financial data. Moreover, the study revealed that using historical prices, technical indicators, and feature engineering could increase efficiency and performance.

In addition, the results of the study demonstrated strong abilities of the LSTM-based hybrid models to learn long-term dependency of stock market dynamics and volatility. Based on the research, it can be stated that hybrid deep learning techniques could help improve forecasting reliability and accuracy.

2.2.17 Automated data processing and feature engineering for deep learning and big data applications

Automated data processing and feature engineering techniques employed in machine learning and deep learning were discussed by Alhassan Mumuni and and Fuseini Mumuni (2024). It should be noted that the authors pointed out that data preprocessing is a crucial stage since unprocessed datasets have missing values, noise observations, inconsistencies, and unwanted information that can adversely affect model results. The study examined data preprocessing stages like data cleaning, imputation, normalization, encoding, and feature selection.

It was found out that preprocessing allows achieving accurate prediction, decreasing computational load, and improving results' credibility in machine learning models. Moreover, it was stated that feature engineering and preprocessing play an important role in improving forecast quality in predictive analytics solutions. Therefore, the literature presented proves that preprocessing procedures, feature engineering, normalization, and cleaning utilized in the project are justified.

2.2.18 Data Cleaning Methods for Improved Machine Learning Model Performance

Ga Young Lee, Lubna Alzamil, Bakhtiyar Doskenov and Arash Termehchy (2021) carried out a study of data cleaning processes and their effect on machine learning efficiency. It was stated by the authors that one of the most critical aspects of predictive modelling is data cleaning since raw datasets usually have missing, inconsistent, noisy, duplicate, and inaccurate data entries. The article discussed different cleaning approaches that include handling of missing values, duplicate deletion, outliers' identification, normalization, and errors correction. According to the authors, adequate data cleaning is vital for improving model accuracy and forecasting performance.

Moreover, it was mentioned that efficient and accurate datasets enable machine learning models to recognize patterns much faster and lower prediction mistakes. The results of the study allowed concluding that data cleaning contributes to improvement of dataset quality and efficiency of models. Thus, the literature review fully confirms the use of data cleaning and preprocessing in the project.

2.2.19 Predict Stock Prices with ARIMA and LSTM

In their comparison of ARIMA and LSTM methods for stock price prediction based on historical market data, Ruochen Xiao, Yingying Feng, Lei Yan and Yihan Ma (2022) found out that LSTM has greater forecasting accuracy compared to ARIMA because of the former's capability to detect sequential dependencies and non-linear relationships in financial datasets.

It was discussed in the article that ARIMA is effective in carrying out linear forecasts but has problems with forecasting highly complex behaviour of stock markets. On the other hand, LSTM models are more capable of detecting trends in markets and learning long-term dependencies in time series.

Thus, researchers argue that LSTM models provide a better forecast of stock prices and are more appropriate for stock markets in general compared to ARIMA models while being harder to apply to less complex cases. This article will serve as the theoretical basis for further work in the current research project.

2.2.20 Financial Time Series Data Processing for Machine Learning

According to Fabrice Daniel (2019), there have been reviews on stock market prediction algorithms, where it has been discussed that the process of collecting and pre-processing data is an integral part of financial prediction. In their work, the researchers noted that stock market data is usually characterized by missing values, noise, duplicates, and other irregularities, which may lower the effectiveness of the prediction.

From the review of literature, through data pre-processing techniques like filling missing values, normalization, elimination of duplicates, and feature extraction, better results can be achieved. The study concludes that data pre-processing enables machine learning algorithms to detect stock market patterns correctly. These findings support data collection and pre-processing used in the current research.

2.3 Research Gap

Even though numerous studies used machine learning and deep learning approaches to predict stock market movements, there are still some gaps left for research. In most cases, researchers paid much attention to enhancing predictive precision, while visualization and interaction reporting tools were neglected.

For this reason, this project will try to close existing research gaps with the combination of Linear Regression, LSTM prediction algorithms, feature engineering, and Microsoft Power BI Desktop dashboards in the framework of one stock analysis and prediction system.

3. Methodology

3.1 Introduction

In this section, the methodological approach used in the project "Stock Analysis and Prediction of Apple Inc. Using Machine Learning and Data Visualization Techniques" is discussed. It describes the processes that were followed from the data acquisition stage through data preprocessing, analysis, evaluation, and visualization.

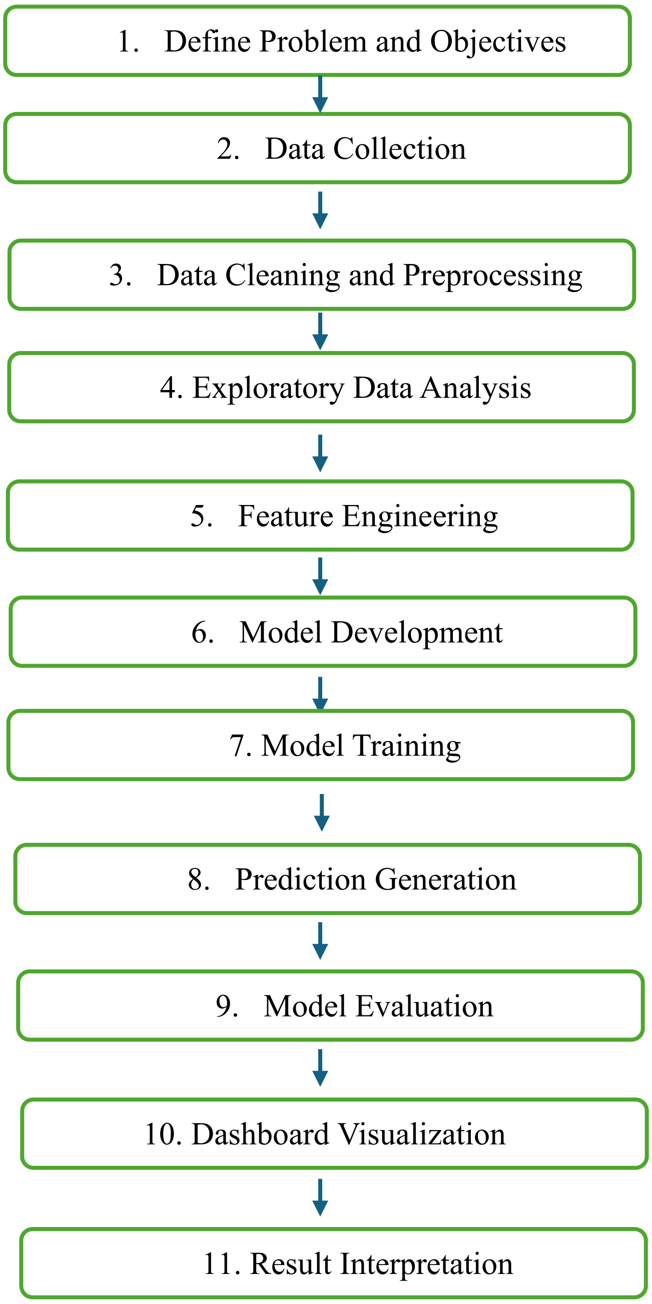
The development of the project was carried out in Python and Microsoft Power BI Desktop. The historical stock data of Apple Inc. were obtained from the website of Yahoo Finance and were analysed using the Linear Regression and LSTM algorithms.

3.2 Research Methodology Framework

This paper’s approach involves a methodical approach where the first step is gathering historical stock data of Apple Inc. on Yahoo Finance, followed by several other steps up until visualization and analysis of predictions.

Data gathering was followed by data preprocessing and exploratory data analysis, which ensured that any cleaning of the dataset and analysis of stock performance would occur. Some of the techniques used during feature engineering were MA20, MA50, and daily returns.

Two different models were considered, where linear regression was used for the machine learning model and LSTM was used for the deep learning model (Ruochen Xiao, 2022). Evaluation metrics such as RMSE and MAE were used to evaluate the models' performance, alongside Power BI dashboard creation. The complete research framework includes:



3.3 Data Collection

3.3.1 Data Source

The main data set for this project contains historical stock market data collected from the website Yahoo Finance. For collecting data, the Python package yfinance was utilized. The purpose of the library is to allow the user to collect stock data from the website without any difficulties.

In particular, the research aims to consider the stock prices of Apple Inc. This corporation is chosen since it is among the largest firms and is widely known across the technology industry. The corporation also holds an important position in world financial markets.

Data relating to the performance of stocks for about ten years was obtained for use in this project. The period in consideration was that between 2015 and 2025 when analysing the performance of stocks.

3.3.2 Data Attributes

The dataset comprises the following variables:

- Date
- Opening price
- High price
- Low price
- Closing price
- Adjusted closing price
- Volume

These data attributes effectively capture the characteristics of stock prices.

3.4 Data Preprocessing

3.4.1 Significance of Preprocessing

Data preprocessing assumes critical significance when it comes to the entire analysis and modelling process (Domingos, 2012). Financial data gathered from the stock market is quite intricate and might have issues like null values, data discrepancies, duplicate rows, and noise. In case these are not dealt with appropriately, it would affect the effectiveness and efficiency of the machine learning models. Hence, data preprocessing becomes essential to enhance the quality of the dataset (Ga Young Lee, 2021).

When it comes to stock market prediction, the quality of data assumes significance since the machine learning algorithms derive their learning through the historical data itself. Hence, the quality of data would impact the behaviour of the algorithm along with the accuracy of its predictions (Daniel, 2019). In this context, data preprocessing enables the dataset to become fit for exploratory data analysis and further implementation of machine learning algorithms.

3.4.2 Data Cleaning

Several data cleaning activities were undertaken to enhance the quality, accuracy, and consistency of the dataset.

The following data cleaning activities were undertaken:

Deletion of duplicate records

Any duplicates in the dataset were found and removed so that there would be no effect on the results or performance of any model used on the dataset.

Correction of data types

The data types of various data attributes were checked for validity and corrected when required. The dates were formatted into datetime type while numeric data types were appropriately formatted.

Validity checking of numeric entries

The dataset was checked for any numeric data such as prices and volumes to confirm that these entries are in numeric form.

Chronological Sorting

Sorting the data based on date in ascending order.

3.4.3 Data Normalization

Data normalization is performed via the use of Min-Max scaling, especially when the LSTM model is implemented (Schmidhuber, 1997). Data normalization entails scaling data between the values of 0 and 1.

3.5 Exploratory Data Analysis (EDA)

To gain information about the characteristics and nature of the stock dataset from history, exploratory data analysis was done. This part of the study helped find the trends and pattern of stocks and markets in the historical period under consideration. This section of the study concentrated primarily on the movements of the stock prices and volumes.

Historical Stock Price Trends

To investigate the nature of past markets and stock performance, historic closing prices of Apple Inc. were studied. The results showed that there was a general positive trend for Apple stock prices through many fluctuations and volatile moments.

Historical Trading Volume

As mentioned before, trading volume was one of the important factors considered in the analysis, as it could help determine the periods of market interest.

Statistical Analysis

The statistical analysis was done using parameters like mean, minimum, maximum, standard deviation, and quartiles. The statistical analysis gave us data related to the stock performance, price fluctuation, and market volatility of Apple Inc.

Trend Visualization

The trend visualization was done by using graphs and parameters like MA20 and MA50 to comprehend the movement of stocks and the trend of the market. These parameters helped smoothen the short-term fluctuations to give us a clear view of the trend.

3.6 Feature Engineering

Feature engineering is a critical phase in machine learning and predicting stock market as raw stock price data might not provide adequate information for accurate prediction (Mahinda Mailagaha Kumbure, 2022). There are often trends and hidden patterns within financial datasets which cannot be observed using original features. As a result, extra features had to be created based on historical stock prices to enhance prediction capabilities.

The goal of feature engineering in this study was to convert raw data from historical stocks into meaningful features that will accurately represent market structure. Through creation of additional variables, stock trends, stock momentum, volatility, and other patterns became better observable for the machine learning algorithms.

There are several features which were generated using the historical data of Apple stock prices including Moving Average 20 (MA20), Moving Average 50 (MA50), return on daily investment, volatility, and lag features (Jinan Zou, 2023).

3.6.1 Moving Average Features Characteristics

Moving average is designed for market trend recognition and smoothing out fluctuations within the stock price range. The following indicators were applied in the current research:

Moving Average 20 (MA20)

Moving Average 50 (MA50)

These features allowed recognizing short-term and long-term market trends.

3.6.2 Calculation of Daily Return

Daily return was calculated to determine percentage changes in the stock price during trading days. The feature allows evaluating stock performance, trends, and volatility according to the closing stock price of Apple Inc.

3.6.3 Volatility Feature Calculation

Volatility is the price variability and uncertainty indicator within a certain period. If volatility is high, it means that there are significant price variations, whereas low volatility indicates stable price ranges. Volatility is calculated as the rolling standard deviation of daily returns.

3.6.4 Lag Features Generation

Lag features allow introducing historical data about stocks to predictive analysis, such as linear regression analysis. Lag1 represents the price of the previous trading day and Lag2 is the price of two previous trading periods.

3.6.5 Significance of Feature Engineering

Feature engineering helped the prediction model to be more efficient by transforming raw information from stock data into meaningful features (Mumuni, 2024). This included the use of features such as moving average, return, volatility, and lag variable features to assist the model in understanding stock behaviours.

3.7 Model Development

3.7.1 Introduction to Model Development

Model development is among the most crucial phases of the project since they develop predictive systems that can analyse previous stock prices and predict their future prices. After performing data preprocessing, exploratory analysis, and feature engineering, machine learning models were built for predicting stocks based on the cleaned and prepared data.

The key idea behind the model development phase is the need to assess whether traditional machine learning models will be able to compete with advanced deep learning models in terms of performance when applied for predicting stock prices (David M. Q. Nelson, 2017). Given the dynamics and the sequential nature of the stock price changes, two distinct forecasting models will be compared in this regard.

The models utilized in the research are as follows:

- Linear Regression (LR)
- Long Short-Term Memory (LSTM)

3.7.2 Linear Regression Model Development

Linear Regression was chosen to be developed as a prediction model due to its simplicity and wide application for forecasting (Wei Bao, 2017). The prediction algorithm uses a linear equation and learns dependencies between inputs and target values.

General Linear Regression equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

Where:

- y= predicted stock price
- β= model coefficients
- x= input variables

3.7.3 Long Short-Term Memory (LSTM) Model Development

Long Short-Term Memory network was chosen as an advanced prediction model as it allows the identification of sequential dependencies in data (Mahinda Mailagaha Kumbure, 2022). The historical stock price of Apple Inc. was normalized through Min-Max scaler and divided into 60-day windows. The neural network consists of one LSTM layer (50 units), Dense layer and trained with Adam optimizer and mean squared error.

LSTM networks contain:

- Memory cells
- Input gates
- Forget gates

- Output gates

These components allow the network to retain important historical information over long periods (Zohren, 2021).

LSTM architecture:

Input Sequence → Memory Layer → Dense Layer → Prediction Output

3.7.4 Data Splitting for Model Training

The dataset was divided into training and testing sets to evaluate model performance.

Split ratio:

- Training = 80%
- Testing = 20%

Training data were used for learning patterns and trends, while testing data were used for prediction evaluation.

3.7.5 Model Training Process

Once the division of data into the training and test datasets was completed, both prediction models were trained on historical stock data of Apple Inc. The Linear Regression model was trained on constructed features like lagged values, moving average, stock returns, and volatility to discover connections between the historical information and stock prices. As for the LSTM model, it was trained on sequential stock price data derived from past trading periods. Training of the LSTM model included normalization of the data along with such parameters as 5 epochs, a batch size of 32 and Mean Squared Error (MSE) loss function. The models gained knowledge of patterns and dependencies within stock price data during training. Training of the models was stopped when the ability to predict prices using the test data was obtained.

3.7.6 Prediction Generation

Post completion of the training phase, Linear Regression and LSTM models were applied to predict stock prices for the testing data set. For this purpose, predictions for the Linear Regression model were made based on the history of data through lag variables and indicators, whereas predictions for the LSTM model were based on the sequence of stock prices. As the LSTM model had been trained on scaled data, an inverse scale transformation was performed on the predictions to obtain the actual value of the stock price predictions. These predictions were later matched against the actual stock prices of Apple Inc.

3.7.7 Comparison of Implemented Models

Both the linear regression model and LSTM were compared to assess their forecasting performances. The simple linear regression model produced easy-to-understand forecasts but could only handle linear associations. On the other hand, LSTM was able to learn and account for market volatility, leading to better prediction performances.

3.8 Model Evaluation

Evaluating the model was done to determine the efficiency and the accuracy in the predictions of both Linear Regression and LSTM Models using the predicted values compared with real stock prices of Apple Inc.

Two evaluation metrics were used in this study:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)

3.9 Power BI Dashboard Development

Interactive dashboards were built using Microsoft Power BI Desktop (Microsoft, 2026) to represent data visualization from stock market data and forecasts. These dashboards were designed to enhance the analytical interpretation and understanding of the stock's performance and predictions results.

Three categories of dashboards were developed in this project. The first dashboard category was the Overview Dashboard that represented general information about the stock performance of Apple Inc. In this dashboard, there were visualizations of stock trends, stock average, min/max values, and filters for interactive analyses based on the stock dates.

The second category of dashboards was the Analysis Dashboard that aimed to visualize and analyse the technical and market aspects of the stocks. In this dashboard, there were visualizations about the moving average, returns, volume, and market trends related to the stock.

Lastly, the third dashboard category was the Prediction Dashboard. This dashboard represented the results obtained by predicting the stock price using the Linear Regression and LSTM models.

3.10 Result Interpretation

Once prediction and visualization were done, results were analysed to identify which one was performing better.

Linear Regression was the baseline model while LSTM was the advanced prediction model.

Accuracy and graphical performance of both models were used in evaluating the best model among the two.

Analysis and prediction results were then displayed using analytical dashboards and prediction visualizations for financial analysis purposes.

3.11 Methodology Summary

In this chapter, we explained the methodology used in this project, which involves collecting and preprocessing the data to building and evaluating our models, finally concluding by visualizing the results using Power BI. This methodology proved to be sufficient for analysing and forecasting the stock price using Linear Regression, LSTM models, and Power BI visualization tools.

4. Implementation

4.1 Introduction

This chapter describes the process of implementing the proposed system to analyse and predict stock prices. The process of implementation includes data collection, analysis, modelling, forecasting, and visualization of results.

Python was chosen as a programming language because of its advanced capabilities in data science, machine learning, and financial modelling. Some of the Python libraries used for the implementation include Pandas, Numpy, Matplotlib, Scikit-Learn, TensorFlow, and Keras. Interactive visualizations were created using Microsoft Power BI desktop.

The implementation was done in several phases, including:

- Data Collection
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Machine Learning Implementation
- Deep Learning Implementation
- Model Evaluation
- Visualizing Results using Dashboards

Data were obtained from Yahoo Finance using the Yfinance Python Library. The dataset was then pre-processed and analysed. Two models for predicting the stock prices were developed using Python:

1. Linear Regression Model
2. Long Short-Term Memory (LSTM) Model

4.2 Data Collection and Dataset Preparation

4.2.1 Data Collection

Data from Apple Inc. from stock exchange was gathered using Python. Yfinance library was chosen for gathering of financial markets datasets from the servers of Yahoo.

The following data was included into the data set:

- Date
- Open price
- High price
- Low price
- Close price

- Adjusted Close price
- Trading Volume

Each of these variables represents data on day-by-day activity on the stock exchange. The data was necessary for time series forecasting conduction and financial analysis.

A long period of historical data gathering was required due to the necessity of receiving enough data for model training. The more historical data is gathered, the better model performance will be, as this data improves forecasting capabilities under different conditions.

```
# -------------------------------------------------
# DATA COLLECTION OF APPLE STOCK FROM YAHOO FINANCE
# -------------------------------------------------

data = yf.download("AAPL", start="2015-01-01", end="2025-01-01") #downloading apple stock price from 01 Jan 2015 to 01 Jan 2025
data.reset_index(inplace=True)

print("Data Loaded:")
print(data.head()) #printing first 5 rows of the dataset
```

```
/tmp/ipykernel_5374/2148492380.py:5: FutureWarning: YF.download() has changed argument auto_adjust default to True
  data = yf.download("AAPL", start="2015-01-01", end="2025-01-01") #downloading apple stock price from 01 Jan 2015 to 01 Jan 2025
[*********************100%***********************]  1 of 1 completedData Loaded:
Price        Date      Close       High        Low       Open     Volume
Ticker                  AAPL       AAPL       AAPL       AAPL       AAPL
0      2015-01-02  24.214888  24.682220  23.776348  24.671145  212818400
1      2015-01-05  23.532721  24.064284  23.346674  23.984549  257142000
2      2015-01-06  23.534941  23.794077  23.173920  23.596956  263188400
3      2015-01-07  23.864946  23.964614  23.632387  23.743129  160423600
4      2015-01-08  24.781889  24.839475  24.075353  24.192741  237458000
```

Figure 1:Data Collection of Apple Stock Price

The downloaded data was stored in a Pandas Data Frame, allowing efficient manipulation and preprocessing.

4.2.2 Data Cleaning

After the dataset was collected, the data had to be prepared and cleaned to ensure that it meets certain criteria and to improve its accuracy. The quality of the dataset is one of the most important factors since accurate predictions depend on the data provided (J. B. Heaton, 2016). Inaccurate or inconsistent dataset will yield wrong results in predictions and other areas of analysis. Therefore, it is essential to preprocess the data before performing any analysis to avoid problems (Berkowitz, 2015).

The following cleaning operations were performed:

- Deletion of duplicate records
- Handling missing values
- Numeric data type validation
- Date conversion
- Chronological sorting
- Index reset

```python
# -------------------------
# DATA CLEANING/PREPROCESSING
# -------------------------


data.reset_index(inplace=True)

new_columns = []
for col_name in data.columns:
    if isinstance(col_name, tuple):

        if col_name[1] == '':
            new_columns.append(col_name[0])
        else:
            new_columns.append(col_name[0]) # We just want the metric name, not the ticker.
    else:
        new_columns.append(str(col_name))
data.columns = new_columns
```

Figure 2: Data Cleaning

```python
# CHECK DATA INFORMATION
print("Initial Data Information:")
print(data.info())

print("\nFirst 5 Rows:")
print(data.head())

# CHECK MISSING VALUES
print("\nMissing Values Before Cleaning:")
print(data.isnull().sum())

# REMOVE DUPLICATES
data = data.drop_duplicates()

print("\nDuplicate Rows Removed")

# CONVERT DATE COLUMN
data['Date'] = pd.to_datetime(data['Date'])

# SORT DATA BY DATE
data = data.sort_values(by='Date')

# HANDLE MISSING VALUES

# Forward fill missing values
data = data.ffill()

# Backward fill remaining missing values (optional)
data = data.bfill()
```

Figure 3: Data Cleaning 2

4.3 Exploratory Data Analysis (EDA)

4.3.1 Trend Analysis

EDA is done on this dataset to understand how the stock market has behaved historically and discover any significant trends present in the data.

Visualization was utilized throughout EDA to find out any trends and fluctuations in the stock prices.

```
# ---------------------------
# 3. EXPLORATORY DATA ANALYSIS
# ---------------------------

plt.figure(figsize=(10,5))
plt.plot(data['Date'], data['Close'])
plt.title("Apple Stock Price Trend")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.show()
```

Figure 4: Trend Analysis Code

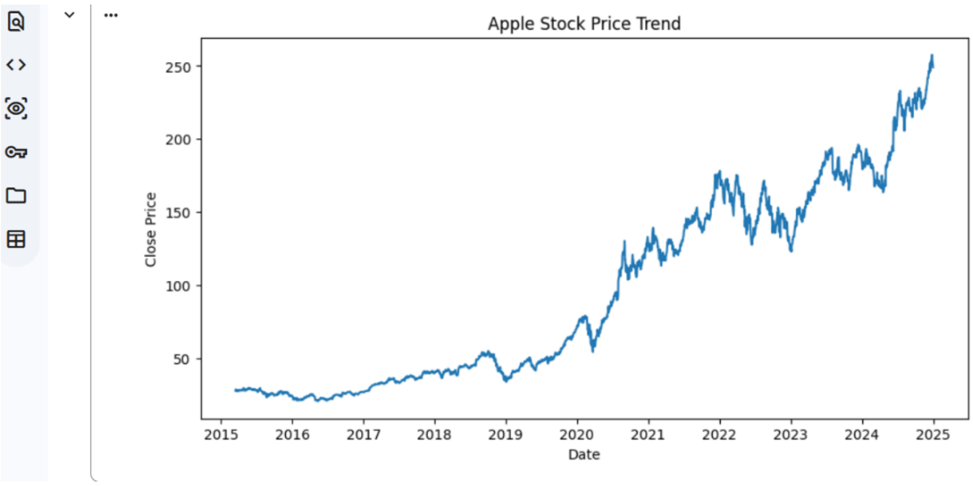


Figure 5: Trend Analysis Visualization

From this exploration, it became clear that there have been some periods of significant long-term growth of Apple stock prices over time.

4.3.2 Volume Analysis

Volume analysis was done to measure investor involvement and market dynamics. Trading volume is a crucial variable in stock market analysis since high trading volume is generally linked to major market occurrences.

```
#VOLUME ANALYSIS
plt.figure(figsize=(12,6))

plt.bar(data['Date'],
        data['Volume'])

plt.title("Trading Volume Analysis")

plt.xlabel("Date")

plt.ylabel("Volume")

plt.show()
```

Figure 5: Volume Analysis

4.3.3 Statistical Analysis

Basic statistical analysis was performed to describe the features of the stock market data set.

The following statistical indicators were obtained:

- Mean
- Standard deviation
- Minimum values
- Maximum values
- Quartiles

```
print(data.describe())
```

Figure 6: Statistical Analysis

4.4 Feature Engineering

4.4.1 Moving Averages

Feature engineering was utilized to enhance prediction performance and generate useful insights from stock data in history.

Two moving averages were calculated:

- 20-day Moving Average (MA20)
- 50-day Moving Average (MA50)

Moving averages can help filter out short-term volatility and identify long-term trends in the market.

```
# 20 Days Moving Average Claculation
data['MA20'] = data['Close'].rolling(window=20).mean()

# 50 Days Moving Average Calculation
data['MA50'] = data['Close'].rolling(window=50).mean()
```

Figure 7: Moving average calculation

4.4.2 Daily Returns

Daily returns were calculated to determine percentage changes in stock prices.

```
# Daily Returns Claculation
data['Returns'] = data['Close'].pct_change()
```

Figure 8: Daily return calculation

Analysis of returns was useful in determining:

- Volatility
- Risk levels
- Daily market fluctuations

4.4.3 Data Normalization

Data normalization was carried out before using the LSTM model since deep learning models are very sensitive to input scale (NAMIN, 2018).

Stock prices were normalized to lie within the range 0 and 1 using Min-Max scaling.

4.5 Linear Regression Model Implementation

4.5.1 Model Development

Linear Regression was chosen as the benchmark model because of its simple nature and ease of understanding. Linear Regression was applied for predicting the prices of stocks belonging to Apple Inc. from their historical prices. The Lag1 feature, which is the closing price of the previous day, served as an independent feature in the regression model, whereas the Close price served as the dependent feature. The missing observations resulting from lagging the series were cleaned prior to building the model. However, due to the presence of nonlinearities, the model performed poorly.

```
# ----------------------
# LINEAR REGRESSION MODEL
# ----------------------

df_lr = data[['Date', 'Close']].copy()

# Lag feature
df_lr['Lag1'] = df_lr['Close'].shift(1)
df_lr = df_lr.dropna()

X = df_lr[['Lag1']]
y = df_lr['Close']

# Train-test split
split = int(len(df_lr) * 0.8)
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

# Train model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

# Predict
lr_pred = lr_model.predict(X_test)
```

Figure 9: Linear Regression model development

4.5.2 Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

This approach ensured that model performance could be evaluated on unseen data.

```
# Train-test split
split = int(len(df_lr) * 0.8)
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

# Train model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
```

Figure 10: Dataset split

4.5.3 Model Training and Prediction

Linear Regression is a technique that can be applied to any set of data where a correlation is present between two variables. The model was developed using historical stock data for Apple Inc., which was divided into 80% training data and 20% testing data. Once the model had been trained, predictions were made from the test data set, and its accuracy was evaluated by comparing it with the real stock price.

```
# Train model
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

# Predict
lr_pred = lr_model.predict(X_test)
```

Figure 11: model training linear regression

4.5.4 Visualization of Linear Regression Results

The results of predictions were graphically shown for comparing real stock prices with predicted prices. Line graphs were utilized for displaying real prices along with the prices predicted with the Linear Regression model. Graphical representation was done to measure how accurately the predicted prices were able to mirror the trend of real stock prices and how different the two sets of prices could be. As observed from the graph, Linear Regression was able to reflect the general rising and falling trend of Apple's stock prices in stable times. During volatile times, however, significant deviations between real and predicted prices could be noticed.

```python
#-------------
# VISUALIZATION
#-------------

# VISUALIZATION ACTUAL VS PREDICTED BASED ON LINEAR REGRESSION
plt.figure(figsize=(14,6))

plt.plot(df_lr['Date'][split:], y_test,
         label='Actual Price')

plt.plot(df_lr['Date'][split:], lr_pred,
         label='Predicted Price')

plt.title('Linear Regression: Actual vs Predicted Stock Prices')

plt.xlabel('Date')

plt.ylabel('Stock Price')

plt.legend()

plt.grid(True)

plt.show()
```

Figure 12: Linear Regression Model Visualization Code

4.6 LSTM Model Implementation

4.6.1 Generation of Sequences

As the LSTM model relies on sequential data, which is essential for making predictions, it is crucial to generate appropriate data sequences that the LSTM model will analyse and predict future stock prices trend based on past data patterns. As such, for the purposes of generating data for the LSTM model, the stock prices of the previous 60 trading days were used as input variables, whereas the stock price for the following trading day was taken as an output variable. Consequently, by using sequences of 60 data points to make predictions, the LSTM model learned to identify both short-term and long-term relationships in the historical stock prices of Apple Inc.

```python
# ----------
# LSTM MODEL
# ----------

# Scale data
scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(data[['Close']])

# Create sequences
def create_sequences(data, seq_length=60):
    X, y = [], []
    for i in range(seq_length, len(data)):
        X.append(data[i-seq_length:i])
        y.append(data[i])
    return np.array(X), np.array(y)
```

Figure 13: LSTM generating sequences

4.6.2 LSTM Architecture

The LSTM architecture included:

- LSTM layers
- Dense output layer
- Dropout regularization

```
# Build model
model = Sequential()
model.add(LSTM(50, return_sequences=False, input_shape=(X_train_lstm.shape[1], 1)))
model.add(Dense(1))

model.compile(optimizer='adam', loss='mean_squared_error')
```

Figure 14: LSTM architecture

The architecture was designed to capture long-term sequential patterns within stock market data.

4.6.3 Model Training

The model was trained on historical stock sequences generated from the past 60 trading days of Apple Inc. stocks. In the process of training, the LSTM network was able to extract sequential patterns and dependencies within the data using the process of weight adjustment and memory state update during the learning process. The model was trained using the Adam optimizer with the mean squared error loss function to achieve low error rates during predictions. This was done through multiple training epochs and batch training processes that allowed the model to enhance learning accuracy during training. The result is an LSTM model that can extract market trend, and volatility patterns present in stock market data for better prediction accuracy (Zohren, 2021).

```
# Train model
model.fit(X_train_lstm, y_train_lstm, epochs=5, batch_size=32)
```

Figure 15: LSTM model training

4.6.4 Predictions and Inverse Scaling

The predictions obtained from the LSTM algorithm were inversely scaled back into actual stock prices after the prediction process. As the input had previously been scaled to Min-Max scaling, the output predictions from the LSTM neural network model had also been scaled to 0 to 1. Inverse scaling was done using the same scaler object used during the training process to return the output predictions into their actual stock price value for interpretation and performance evaluation. Inverting the output predictions would enable the predictions to be compared to the actual stock prices of Apple Inc. The inverse scaling process was essential since it made the final predictions useful and interpretable.

```
# Predict
lstm_pred = model.predict(X_test_lstm)

# Inverse transform
lstm_pred = scaler.inverse_transform(lstm_pred)
y_test_actual = scaler.inverse_transform(y_test_lstm)
```

Figure 16: LSTM prediction and inverse scaling

4.6.5 Visualisation of LSTM Prediction Results

The prediction results were illustrated in the form of graphs for comparing with the actual price of the stocks to assess the efficiency of the model in predicting the future movement of stocks. A line graph was employed in presenting the stock prices, which allowed a comparison between the predicted and actual stock prices. It was found from the illustration that the LSTM prediction model was able to follow the market dynamics, as well as capture the short-term and long-term changes in the prices of Apple Inc. stocks. As compared with the Linear Regression model, the predictions made by the LSTM model possessed less lag time and followed the trends of stock prices more smoothly.

```
# VISUALIZATION BASED ON LSTM MODEL
# --------------------------------

plt.figure(figsize=(14,6))

plt.plot(
    data['Date'][-len(y_test_actual):],
    y_test_actual,
    label='Actual Price'
)

plt.plot(
    data['Date'][-len(lstm_pred):],
    lstm_pred,
    label='Predicted Price'
)

plt.title('LSTM: Actual vs Predicted Stock Prices')

plt.xlabel('Date')

plt.ylabel('Stock Price')

plt.legend()

plt.grid(True)

plt.show()
```

Figure 17: Visualisation of LSTM Prediction Results

4.7 Model Evaluation

4.7.1 Evaluation Metrics

The forecasting models created in the current research project have been assessed using statistical criteria to evaluate their performance and accuracy. Evaluation is a vital step when implementing a machine learning and deep learning model because it allows one to estimate the effectiveness of a particular algorithm based on its results on testing data. In this study, the models have been tested by comparing the forecasted stock price with the real price of Apple Inc. shares.

The following metrics have been used to evaluate the models:

- RMSE
- MAE

```
# -----------------
# EVALUATION METRICS
# -----------------

lstm_rmse = np.sqrt(mean_squared_error(y_test_actual, lstm_pred))
lstm_mae = mean_absolute_error(y_test_actual, lstm_pred)

print("LSTM RMSE:", lstm_rmse)
print("LSTM MAE:", lstm_mae)
```

Figure 18: Evaluation matrix

4.8 Implementation of Power BI Dashboard

4.8.1 Dashboard Development

Interactive dashboards were created with Microsoft Power BI Desktop for visualization of the trends of the stock market, technical analysis, and predictions of stock prices made throughout the project period (Zohren, 2021). While creating dashboards, the emphasis was placed on making complex analytical data more visually understandable and interactive for users. Different types of visualization methods, such as line charts, bar charts, KPI cards, slicers, and comparisons, were employed for enhancing the interpretation of trends in the stock market and performance of models created. Information that included closing prices, moving averages, trading volumes, daily returns, and actual-predicted comparisons of stock prices was presented in the dashboards. Also, interaction capabilities allowed for further exploring and analysing the data from different perspectives within different time intervals.

4.8.2 Dashboard Pages

The Power BI dashboard consisted of three major pages:

Overview Dashboard

Included:

- KPI cards
- Stock trend line charts
- Date slicers

Analysis Dashboard

Included:

- Volume analysis charts
- Returns analysis charts
- Moving average visualizations
- Date Slicers

Prediction Dashboard

Included:

- Actual vs predicted stock prices
- Error analysis
- RMSE and MAE indicators
- Date Slicers

The dashboards enabled interactive analysis and easier understanding of stock market behaviour.

5 Results and Evaluation

5.1 Overview

In this chapter, the findings of the application of machine learning and deep learning approaches to predict the prices of stocks of Apple Inc. company will be presented. The key aim of this chapter is to assess the success of the developed algorithms in predicting stock price dynamics based on historical data. In addition, it will be explored how successfully these models can forecast the future price by evaluating their advantages and disadvantages.

There are two prediction models applied in this research project:

1. Linear Regression (LR)
2. Long Short-Term Memory (LSTM)

The testing of both models was performed using testing data not used for training. The following performance metrics were calculated to assess models: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE). Also, some visual analysis of the differences between predicted and real values of stock prices was made.

In this chapter, not only numerical results of predictions will be discussed, but also how successfully each algorithm coped with the task of modelling stock prices in volatile markets will be analysed.

5.2 Experimental Design

A ratio of 80:20 was employed for the training and testing datasets. The historical stock dataset was divided into two parts such that around 80 percent of the stock data was used for model training while the other 20 percent of stock data was used for testing purposes. In this way, the models were trained on enough historical data as well as tested on unknown data to evaluate the ability of the models to generalize.

The experiment was conducted using historical stock price data of Apple Inc. collected from Yahoo Finance. The dataset included stock variables such as:

- Open price
- High price
- Low price
- Close price
- Volume
- Moving averages
- Returns

The models used for stock price prediction are as follows:

Linear Regression (baseline model)

LSTM (deep learning model)

The performance of both models was analysed using the following metrics:

Root Mean Square Error (RMSE)

Mean Absolute Error (MAE)

5.3 Results of Linear Regression Model

5.3.1 Model Performance

Linear Regression Model was used as the baseline machine learning algorithm in predicting the future prices of the stock based on the historical stock price data. This algorithm helped to recognize the overall increasing and decreasing trends of Apple Inc. stock prices and generated accurate predictions under normal circumstances in the market.

The predictions were fairly accurate while market conditions remained constant, as they generally matched the actual trends in the stock prices. But they were inaccurate during volatile market conditions since the algorithm assumed the presence of a linear relation between the input parameters and the output (target) stock prices.

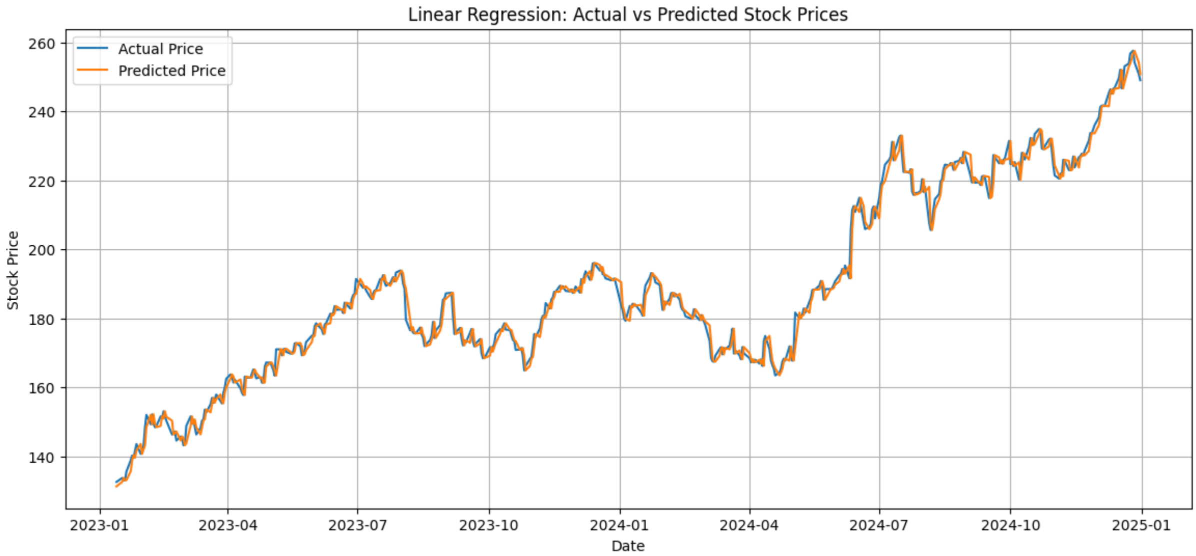


Figure 19: Linear regression model performance

5.3.2 Measures of Errors

The predictive accuracy of the Linear Regression model was assessed by using RMSE and MAE measures.

Higher RMSE figures were indicative of more significant forecast errors made by the model at moments when there were sudden changes in the stock market. Similarly, high MAE figures were indicative of significant deviations in actual versus predicted prices in general.

The predictive accuracy of the model was adequate, but it was not sufficient considering the complicated and non-linear nature of stock market data.

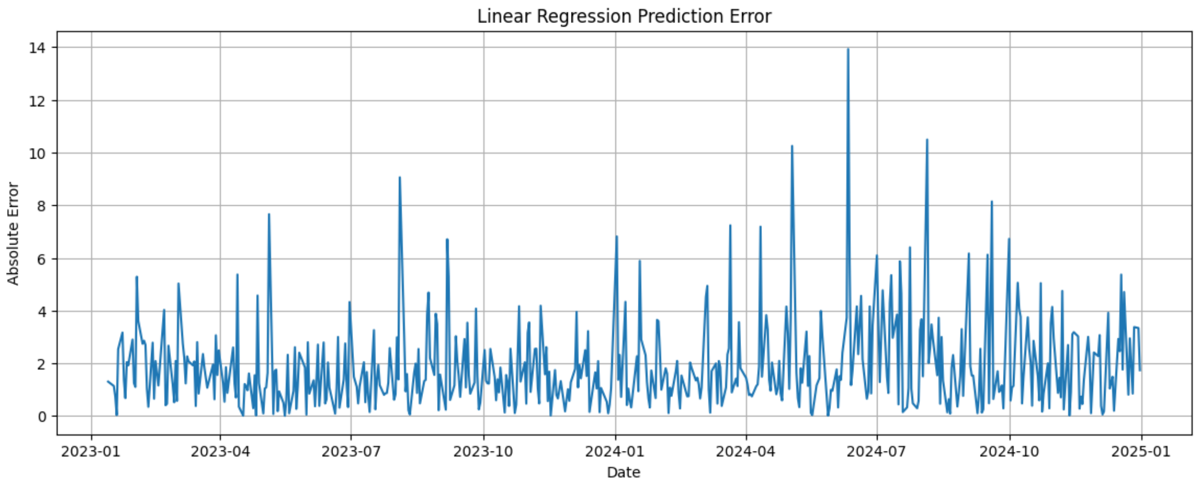


Figure 20: Linear regression error measure

5.3.4 Critical Evaluation

The following are the benefits offered by the Linear Regression model:

- Easy to use
- Quick training time
- Cost-effective computations
- Interpretability

On the other hand, the model exhibited some weaknesses:

- Could not detect non-linear trends
- Poor accuracy in volatile situations
- Linear-based assumptions

Stock markets are quite dynamic and are affected by numerous variables. This means that the simplicity of the model did not enable accurate predictions. Regardless, the model acted as a useful benchmark to test the accuracy of the LSTM deep learning model.

5.4. Results of LSTM Model

5.4.1 Performance of Predictions

It is clear from the results that the LSTM model performed significantly better than the Linear Regression model due to the advantage it had in processing the sequence of input data and maintaining information through memory units (Jinan Zou, 2023). In contrast to other approaches used in machine learning, the LSTM model could learn the long-term dependencies and patterns in the stock price data of Apple Company. Thus, more accurate predictions could be made.

Predictions made using the LSTM model were close to the real values, even at the times when there was a certain level of volatility on the market. Moreover, the predictions were made in line with both the general up and down trends of the market and the small deviations from them. This model performed better in comparison with the Linear Regression one.

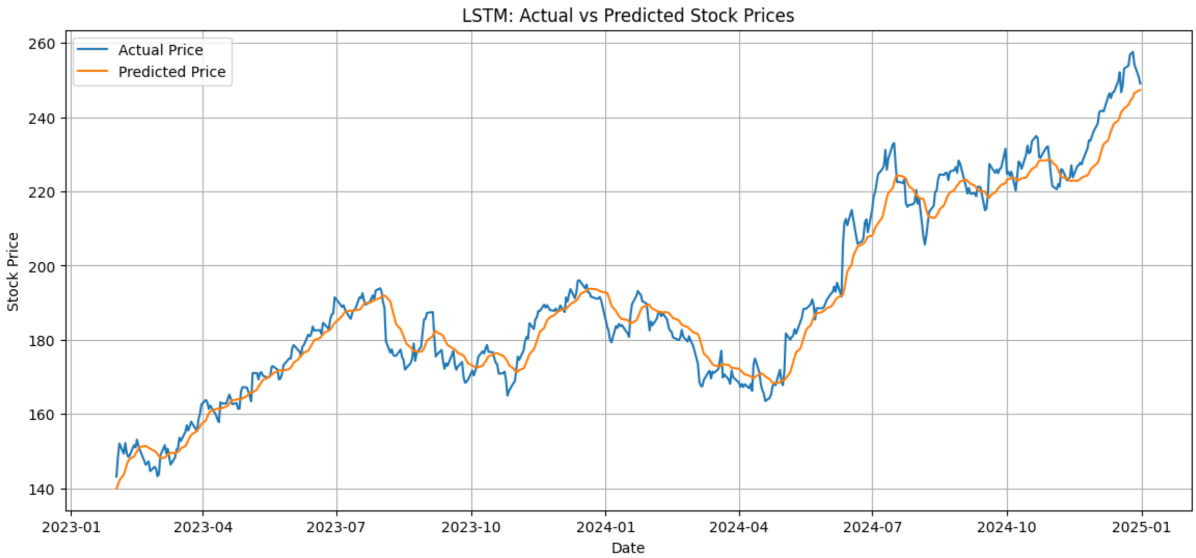


Figure 21: LSTM model performance

5.4.2 Error Measures

The LSTM model was able to yield lower RMSE and MAE scores, demonstrating better performance in the process of future stock prices prediction when compared to the Linear Regression model. This is since the lower values of these error measurements mean that there is a higher correlation between predicted and real values.

Lower RMSE = fewer big errors

Lower RMSE means that the number of large prediction errors has been minimized in the process of prediction made with the help of the LSTM model.

Lower MAE = general improvement

The decreased value of MAE means that on average the difference between real stock prices and those predicted with the help of the model became smaller.

Overall, all these measures of performance showed that the LSTM model outperformed the Linear Regression model.

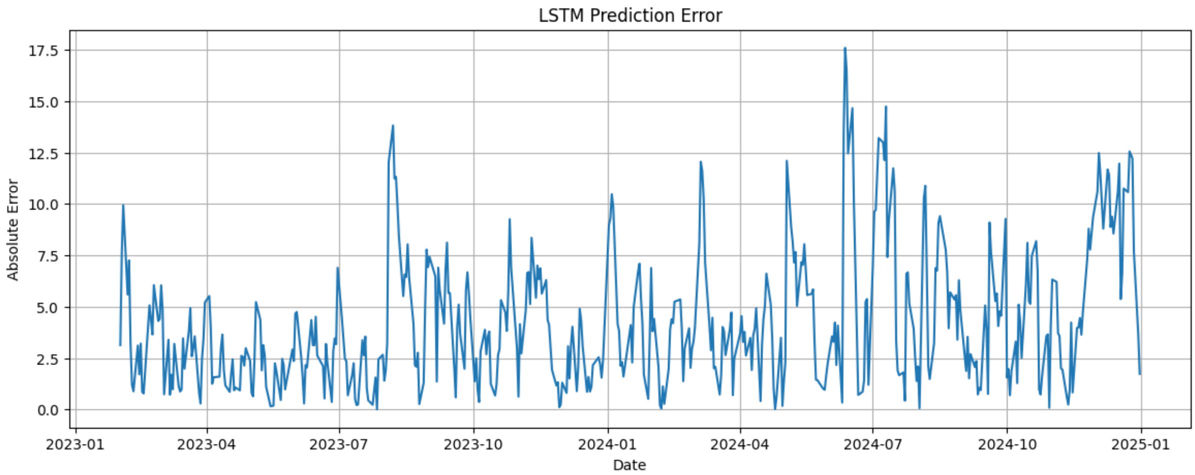


Figure 22: LSTM error measures

5.4.3 Graphical Analysis

The graphical analysis, which is another method of model comparison, confirmed the better results produced by the LSTM model. By analysing how the real and predicted stock prices appeared in the visualizations, one could observe that the graph of predicted prices resembled the actual market dynamics.

The graphical analysis made it possible to make the following observations:

- **High correlation between predictions and real numbers**
- **Better trend detection**
- **Fewer lags than Linear Regression**

It can be concluded that sequential learning played a key role in obtaining the results.

5.4.4 Critical Evaluation

The increased efficiency of the LSTM model was due to its capacity to incorporate sequentially and memories when learning from examples. The difference between the LSTM and linear regression models was that the former did not have any linear constraints and could thus effectively learn from non-linear relationships between different variables in the dataset (Yann LeCun, 2013).

With the help of memory cells, the model was able to remember important information from past sequences for a longer period and was thus able to identify dependencies in changes in stock prices. In this way, the LSTM network was more adaptable to changes in market trends and could predict them.

5.5 Comparison of Models

5.5.1 Comparison of Model Effectiveness

The LSTM model outperformed the Linear Regression model in all parameters. Therefore, deep neural networks are better suited for predicting time series.

Evaluation Factor	**Linear Regression**	**LSTM**
Prediction Accuracy	Moderate	High
Trend Following	Basic	Strong
Volatility Handling	Weak	Better
Computational Cost	Low	High
Sequential Learning	No	Yes
Non-linear Learning	No	Yes

5.6 Effect of Feature Engineering

Feature engineering was instrumental in enhancing the predictive capabilities of the model.

Specifically:

- Moving averages enabled trend detection
- Volatility indicated risk
- Returns offered short term trend information

5.7 Market Trends Analysis

From the analysis, it has been possible to determine some market trends in relation to Apple stock prices such as:

- Long term upwards growth

- High levels of volatility
- Trends affected by market situations

5.8 Power BI Dashboard Results

5.8.1 Overview Dashboard Results

An Overview Dashboard was designed to give a broad overview of the historic stock performance of Apple Inc. The key performance indicator cards illustrated vital details like Average Price, Max Price, Min Price, and Total Trading Volume. These metrics offered a brief look into the performance of the stock market for the chosen period.

The Closing Price graph illustrated the general trend of increasing stock prices at Apple Inc. There were times of volatility and market corrections. Moving averages were used to make trend analysis simpler by minimizing short-term variations and focusing on long-term trends. Date filters were used to enable stock performance analysis for other time frames. In summary, the Overview Dashboard effectively simplified historic stock data into visual information.

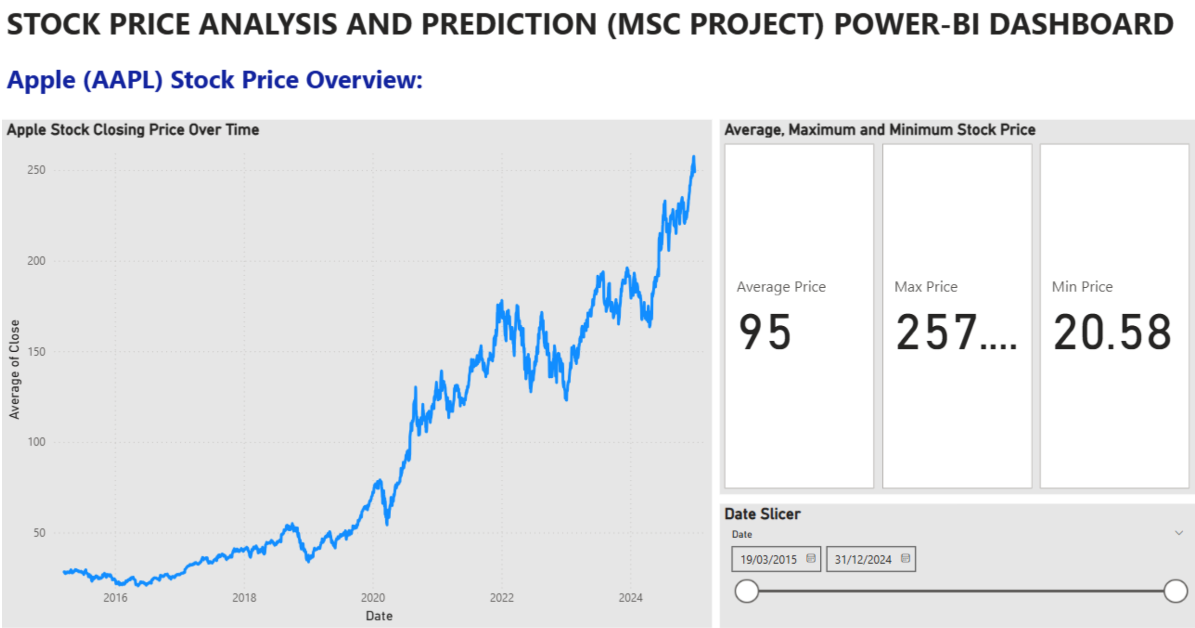


Figure 23: Overview PowerBI dashboard

5.8.2 Analysis Dashboard Results

Analysis Dashboard concentrated more on technical and statistical evaluation of behaviour of stock market. Charts such as trading volume charts, returns analysis graphs and moving average graphs were used for evaluating stock market behaviour and volatility.

From the graph of volume, we could see that high volumes are linked with big moves in the stock market. This meant a lot of participation from investors in the stock market when something major happened in the stock market. Similar was the case with returns analysis where it was seen that daily returns of stocks were quite volatile.

Moving average charts gave information about the long run and short run directions in the market. Crossovers of moving averages MA20 and MA50 indicated market changes and changing investor sentiments.

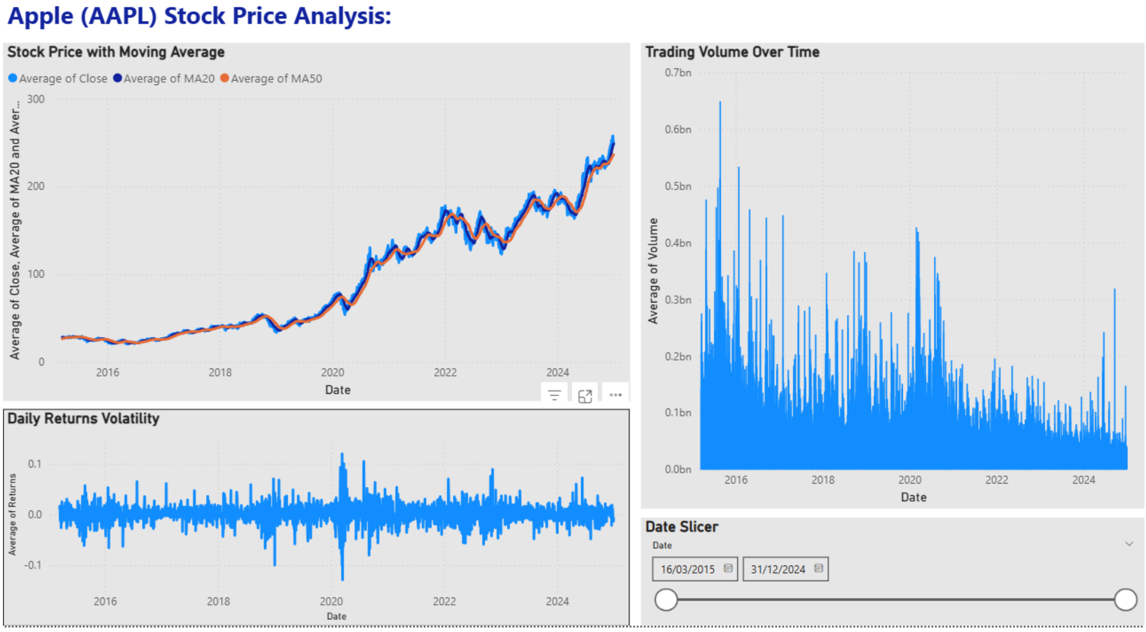


Figure 24: Analysis PowerBI dashboard

5.8.3 Prediction Dashboard Results

In the Prediction Dashboard, the forecasts made by the Linear Regression and LSTM models have been shown graphically. The primary chart displayed actual vs. predicted stock prices through an interactive line chart.

It is evident from the dashboard's outputs that the predictions made by the LSTM model were much more accurate than the predictions made by the Linear Regression model. This was evident from the fact that the LSTM model's prediction line closely matched the actual line on the graph, whereas the linear regression model's prediction line was somewhat erratic.

Moreover, other graphs were also included in the dashboard that showed prediction errors in terms of RMSE and MAE. Other KPIs like RMSE and MAE also verified that the LSTM model outperformed the Linear Regression model regarding prediction accuracy.

Therefore, it can be concluded that visualization is essential in evaluating predictive analytics' results effectively.

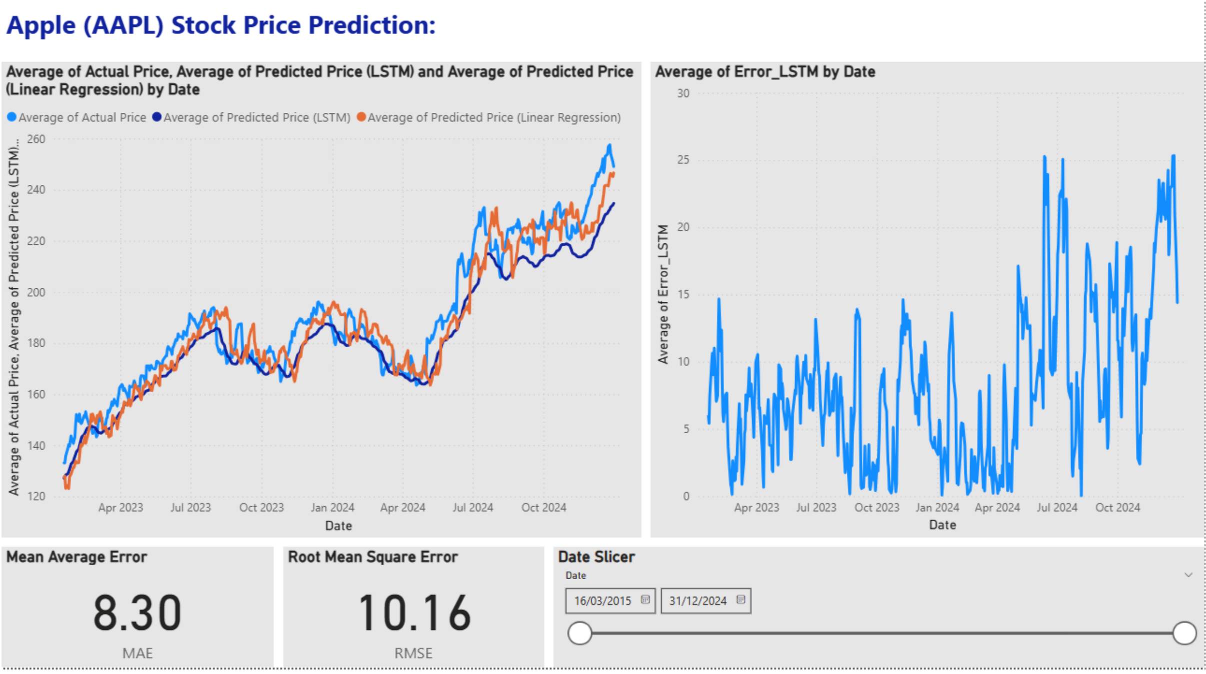


Figure 25: Prediction PowerBI dashboard

Prediction of Stock Price Trend in Future:

The LSTM prediction line is much closer to the actual line compared to the Linear Regression prediction line, which shows a better performance in capturing trends within the data set. Since the current trend is going upwards and the prediction errors are relatively low (MAE=8.30, RMSE=10.16), then we can predict that Apple stocks will continue having an upwards trend in the foreseeable future despite the occasional fluctuations in the stock market. Consequently, the forecast generated by the dashboard is bullish for future stock performance.

5.9 Discussion of Results

Machine learning algorithms can be successfully applied for predicting stock prices. Nevertheless, there are several key observations which should be considered:

Model Performance

A. LSTM outperforms because of its capacity to detect dependencies
B. Linear Regression suffers from simplicity

Restrictions on Prediction

Stock prediction is still associated with uncertainty caused by:

- Volatility
- External factors (news, political issues, economic events)
- Randomness

Overfitting and Generalization

There is a possibility that the LSTM model could be prone to overfitting the training data if appropriate measures are not taken. It is critical to ensure the generalization to the unseen dataset.

Reliability of Results

Even though both proposed models generate accurate predictions, their outcomes cannot be considered completely reliable. Models should be used as auxiliary means of analysis, but not for making decisions about investments.

5.11 Limitations of the Study

While there have been successful predictions, the following limitations can be observed:

1. Insufficient data features
2. The absence of exogenous variables (sentiment news, macroeconomic indicators)
3. Costly computational processes
4. Temporary forecasts

5.12 Recommendations for Improvement

Some potential improvements that could be made in future include:

- Machine learning algorithm with sentiment analysis capability
- Multi-step modelling approach
- Implementing prediction systems in real time

5.13 Summary

The findings of both models (Linear Regression and LSTM models used for Apple stocks prediction) were discussed in this chapter. Specifically, the Linear Regression model was able to predict general market movements but showed poor results in times of market turbulence because of its linear nature. In turn, the LSTM model outperformed the regression algorithm by being capable of recognizing sequential behaviour and non-linearity. It is supported by low RMSE and MAE results as well as by the visualizations of the prediction dashboard. Moreover, the prediction dashboard showed that there is a positive trend expected for the future performance of Apple stocks because the predictions made with the help of the LSTM model were very close to actual stock movements.

6 Discussion

6.1 Overview

In discussion section, an evaluation of the results obtained after applying the machine learning and deep learning models for forecasting the price of stocks of Apple Inc. is discussed. In other words, this chapter emphasizes on critical analysis of the findings that were found in the previous chapter.

The main emphasis in this part would be on the performance of the Linear Regression Model as well as the LSTM models, along with feature engineering and visualization. In addition, comparisons will also be made with previous research works.

Lastly, the problems faced while forecasting stock prices and how these can be overcome will also be explained.

6.2 Discussion of Findings

The findings from this project indicate that machine learning and deep learning techniques can be effectively utilized for stock prices trend forecasting. Both the Linear Regression and LSTM models have been effectively used to study past stock prices and forecast future stock prices patterns for Apple company. However, it is notable that there was a difference in the predictive accuracy of the two models.

Linear Regression model can effectively identify stock price trends and generate accurate forecasts when the market conditions are stable. This clearly suggests that historical stock prices exhibit linear relationships, which can be leveraged for forecasting. Nonetheless, the model did not work effectively under volatile market conditions since linear relationships were assumed.

On the other hand, LSTM model generated better prediction and more accurate forecasts compared to the Linear Regression model. This is attributed to the ability of the model to capture sequential relationships and adjust its prediction based on changes in stock market behaviour using memory cells. This is evident from the lower RMSE and MAE values. Lastly, according to the prediction result the price trend will be upward despite of having a few fluctuations.

6.3 Relationship Between Findings and Research Questions

The current research project had multiple research objectives. These questions can now be assessed considering the findings:

6.3.1 Effectiveness of Machine Learning Models

As seen from the findings, the machine learning models used are effective in predicting stock price trends. The two models used were able to predict price trends that follow the general pattern observed in the actual data. Nevertheless, there was variation in the accuracy between models.

From this, it can be concluded that although machine learning methods are highly valuable when analysing stocks, the selection of models is crucial.

6.3.2 Comparison Between Models

Yet another main aim of the research was to assess how traditional machine learning techniques and deep learning approaches compared with each other.

It was evident that:

- Linear regression had a satisfactory performance in stable markets
- LSTM worked better in cases of volatility and sequentially
- LSTM resulted in smaller errors of predictions
- Deep learning had better trend following ability

All this indicates that forecasting of stocks is greatly facilitated by techniques capable of learning dependencies between the input parameters.

6.3.3 Importance of Visualization

Another objective of this project was to include visualization techniques into the analysis and forecast of stock markets.

Using Microsoft Power BI Desktop for creating dashboards helped enhance both interpretation and representation of results of the analyses conducted. Visualization techniques enabled transforming complicated financial information into visual representations to the charts and KPIs.

Thus, this project successfully proved the significance of integration of predictive analytics and visualization tools.

6.4 Critical Analysis of Models' Performance

Although the results suggest that LSTM is more precise in predicting stock prices, there are some critical aspects regarding the model's performance that need to be discussed.

6.4.1 Advantages of LSTM

The LSTM model was advantageous over the Linear Regression model on a few accounts.

Firstly, it possessed the following key strengths:

- Capability to recognize sequential patterns
- Effective processing of time-series data
- Greater prediction accuracy
- Superb flexibility during turbulent times
- Superior trend following capabilities

Secondly, due to the use of memory cells, the LSTM neural network could retain data history, which enhanced the network's capacity to understand patterns in the stock markets.

6.4.2 Disadvantages of LSTM

Although the LSTM model exhibited excellent performance, there were also some constraints that accompanied it.

They included:

- High computational complexity
- Increased training time
- Larger datasets requirement

- Tuning hyperparameters
- Overfitting threat

The machine learning algorithm was computationally intensive compared to the deep learning algorithm. The LSTM architecture involved higher computational complexity and required larger datasets.

6.4.3 Shortcomings of the Linear Regression model

Despite being efficient and straightforward, the predictive performance of the Linear Regression model was still quite limited.

The model largely depended on the temporal consistency of the preceding and subsequent stock prices. This led to the predictions appearing more stable than they would during volatile periods in the markets.

However, the model served as an excellent benchmark for comparison and testing the efficacy of deep learning approaches.

6.5 The Importance of Feature Engineering

Feature engineering was crucial in improving the ability of machine learning and deep learning models to make predictions.

This was achieved through the introduction of new features including:

- Moving averages
- Returns
- Lagged values
- Volume indicators

These features helped to add more information to the model.

Using moving averages helped to smoothen the short-term fluctuations while returns helped to identify market volatilities and daily performance of the stocks.

6.6 Challenges of Stock Price Prediction

Market Volatility

The stock markets are very volatile and prone to rapid changes. The sudden change in the economy, reactions of investors, political happenings, and world events may significantly impact the price of stocks in a relatively short period.

This aspect is very problematic for predictions because the market behaviour is not always predictable.

Effects of External Influences

This project was based mainly on past stock market information and technical indicators. Yet, there are many other influences that affect the price of stocks, including:

- Economic indicators
- News sentiments
- Reactions on social media platforms
- Interest rates
- Corporate communications

Ignoring these variables may decrease the effectiveness of the predictions.

Presence of Noise in Financial Datasets

Many financial datasets have many noises and unpredictable patterns. These aspects may affect the accuracy and reliability of the predictions.

6.7 Comparison with Existing Literature

The conclusions drawn from this study are in line with those made by other researchers, who claim that deep learning algorithms outperform the classical machine learning approaches in predicting the stock market. Indeed, earlier research demonstrates that LSTM-based models work well in predicting time series because they can identify sequences and understand long-term dependencies inherent in financial data. Moreover, deep learning methods prove to be more efficient in dealing with financial market volatility than traditional statistical and machine learning models.

Thus, the conclusions drawn from the current research confirm the efficiency of using deep learning in stock market forecasting while identifying the limitations of such prediction associated with market volatility and uncertainty, as well as the non-linearity of financial data. Specifically, the LSTM-based approach worked better than the Linear Regression one, proving its superior capability to track the stock market dynamics.

6.8 Practical Implications

The current research offers various practical applications as described below:

Investors

The investors can use the predictive analysis tools for:

- Monitoring stock performance
- Making investment decisions
- Analysing the market behaviour
- Evaluating volatility

However, predictions cannot serve as the only basis for making financial decisions since stock markets are unpredictable.

Financial Analysts

Financial analysts can utilize machine learning techniques to enhance the accuracy and efficiency of the traditional prediction methods.

Organizations

- Firms/financial institutions can employ predictive systems for:
- Analysing the market trends
- Planning strategically
- Making financial decisions
- Creating an automated forecasting system

6.9 Limitations of the Project

Following limitations should be noted:

1. Limited Features Set

   Only historical price data used

2. Lack of External Sources

   No news sentiment nor any economic indicators were considered

3. Model Restrictions

   Hyperparameter optimization limited by time and resources

4. Prediction Range

   Short-term predictions considered only

6.10 Recommendations for Future Research

Considering the results and limitations of the study, the following suggestions can be made for further research. The future research may use sentiment analysis based on news articles and social media data to achieve greater prediction accuracy. Moreover, more sophisticated models such as those built on a transformer architecture can be used to increase the effectiveness of prediction. Finally, the improvement in feature engineering and real-time stock prediction can help achieve better results in the stock market forecasting.

6.11 Reflection on Methodology

The methodological approach applied to this research effectively blended:

- Data analysis
- Machine learning
- Deep learning
- Visualization methods

The implementation of predictive modelling using Python coupled with Power BI dashboarding was an effective and applicable approach to analysing the stock market.

It was also shown that:

- Data pre-processing

- Feature engineering
- Sequential learning
- Interactive visualization

These elements are vital for such a study.

6.12 Conclusion

In conclusion, discussion chapter gave a critical analysis of the project findings with emphasis on the merits and demerits of the used model. The findings show that although machine learning models including the use of LSTM model enhance the accuracy of predictions, predicting the stock market is a challenging area.

It is worth noting that the adoption of predictive modelling together with visualization tools increases the overall effectiveness of the whole project.

7. Conclusions and Future Work

7.1 Introduction

This part offers an overview of the results obtained in the current research project through the analysis of the realization of the research objectives, as well as the strengths and weaknesses of the current study. This chapter also provides the potential directions for further research that could enhance the performance of stock market prediction models. Finally, this chapter provides a summary of the whole research process, including data preparation and modelling, and assessment of Machine Learning and Deep Learning techniques.

7.2 Summary of the Project

The primary goal of this project is to analyse and forecast stock prices of Apple Inc. using machine learning and deep learning approaches with visualization through business intelligence tools. Stock market historical data were acquired using Python from Yahoo Finance which include features like open price, closing price, high price, low price, and volume.

Exploratory Data Analysis (EDA) was conducted to determine the behaviour of stocks, volatility, and market conditions. Feature engineering methods were used by creating new features like Moving Average (MA20 and MA50), Returns, and Volatility to enhance the accuracy of predictions.

Two prediction models were created in this project which are Linear Regression for machine learning approach and Long Short-Term Memory (LSTM) for deep learning approach. The performance of the models was assessed using metrics like RMSE and MAE. Lastly, dashboards of the predictions and analysis results were created using Microsoft Power BI Desktop.

7.3 Achievement of Objectives

The project successfully achieved all objectives that were outlined right at the start of the experiment. Every step in the project was conducted in a logical order, contributing to the achievement of the main objective of predicting stock prices.

Data Collection and Preprocessing

Data collection was successful in collecting historical data from stock markets. Preprocessing of data entailed tasks such as handling of missing values, duplicate removal, data type conversion, chronological sorting of data, and preparation of data for machine learning analysis. Successful feature engineering was also done.

Conducting EDA

The Exploratory Data Analysis was successfully carried out using statistics and visualization tools. Insights obtained from the analysis provided valuable information regarding the behaviour of historical stock prices, the behaviour of volatilities, and relationship among various variables.

Model Development

Both linear regression and LSTM models have been successfully developed through Python. This entailed developing, training, and optimizing the models using stock history data to make future stock predictions.

Model Evaluation

The predictive models have been successfully evaluated through standard model evaluation measures such as RMSE and MAE. From the evaluation findings, the LSTM model outperformed the linear regression model in prediction accuracy and trend following ability.

Data Visualization

Through Microsoft Power BI Desktop, interactive dashboards have been developed successfully. They made stock market trends and predictions easier to understand through their visualization capabilities.

7.4 Significant Findings

There have been many significant findings made during the process of implementing and evaluating the machine learning models in the project.

Effectiveness of Machine Learning Models

It was discovered that machine learning models can predict and analyse the pattern of the stock markets to a reasonable degree of accuracy. The simplest model in terms of complexity, which is Linear Regression, can provide a general pattern while deep learning models offer high accuracy.

Importance of Using the LSTM Model

Deep learning methods like the LSTM model have been found to be superior to Linear Regression because of their abilities to learn from time-series data and use memory cells to maintain the information about past values.

Importance of Feature Engineering

The process of feature engineering is very significant for machine learning tasks, especially in the field of finance. It allows recognizing patterns in the data and thus improve prediction accuracy by using specific features like Moving Averages, Returns, and Volatility indicators (Sidra Mehtab, 2020).

Importance of Visualization

Visualized representations of analysis results made it easier to interpret outcomes by means of dashboards provided by Microsoft Power BI Desktop tool.

Prediction of Analysed Stock Price Trend

The predicted line based on the LSTM model is much nearer to the actual stock price line than the Linear Regression model, proving its superiority in recognizing trends and patterns in the data. As the general trend in the stock is upward and the errors in predictions are still quite low, it is likely that the prices of Apple stocks will keep on moving upwards even in the coming period despite some ups and downs in the market. Hence, a bullish prediction can be made regarding the performance of the stock.

7.5 Critical Analysis of the Study

Despite the successful completion of the project and achievement of the stated aims and objectives, certain limitations and difficulties have been observed.

Weaknesses Linked to the Selected Data Type

Only data linked to the stock price history and technical indicators was considered while conducting the research. Factors such as news, sentiments, politics, and economic indicators were not accounted for. They may affect stock prices and lead to an increase in prediction accuracy.

Weaknesses of the Used Method

Although LSTM produced more accurate predictions than a linear regression model, deep learning algorithms required more powerful computing resources and complex hyperparameters adjustment. Moreover, they usually lack transparency.

Forecast Uncertainty

There is no way to predict stock market activity since the stock market is characterized by unpredictability. Unexpected economic events, global crises, and changes in sentiments may impact it negatively and unpredictably.

Scope of Work

During the work, stock prices of Apple Inc. were analysed. Although this decision helped conduct research effectively, it limited its generalizability.

7.6 Contribution to the Field

The current project makes several contributions to data analytics and financial forecasting:

- Introducing machine learning and deep learning algorithms to the stock market forecasting process
- Comparing traditional machine learning and innovative deep learning methods
- Combining predictive analytics with interactive visualization dashboards
- Highlighting the necessity of feature engineering and data preprocessing
- Implementing Python and Power BI software tools for practical financial analytics purposes

7.7 Practical Implications

The findings of the study provide multiple practical implications for financial analytics and business decision-making.

Investors

Investors can apply the knowledge gained through the presented project to analyse stock market trends and make investment decisions based on the acquired information. At the same time, predictions should not serve as the only source of guidance for making investment decisions.

Financial Analysts

Financial analysts can incorporate machine learning algorithms into conventional financial analysis methods to enhance the accuracy of forecasting and automate the process of analysing financial data at scale.

Companies

Companies can apply predictive systems to monitor market behaviour, assess investment options, and design an appropriate approach to strategic financial planning.

7.8 Recommendations for Future Research

Several recommendations for future research can be made considering the findings and limitations of this paper.

Inclusion of Additional Data

Additional data that could be included in future studies may include, but not be limited to:

- Sentiment analysis of news articles
- Information from social media platforms
- Various economic indicators
- World market dynamics

Development of More Complex Models

For improving prediction accuracy, future research could consider applying the following models:

- Models based on transformer architecture
- Hybrid machine learning models
- Attention-based neural network models

Real-Time Prediction

To make the stock price prediction system fully functional, researchers could consider implementing a real-time approach to stock price prediction.

Predicting Stock Price Changes for Several Companies

Predicting stock price changes for several companies at once is another promising area to explore in future research.

Enhancement of Feature Extraction Technique

Future work could be focused on finding new financial indicators and features that can be used to improve the prediction system.

7.9 Conclusion

Initially, the ARIMA model was considered in proposal for my project. However, during the implementation process, it was seen that the data on the stock prices of Apple was volatile, shown sequence dependency, and had nonlinear behaviour. I have gone through the previous literatures of comparing ARIMA and LSTM and found LSTM performed better. moreover, the LSTM model was better suited to deal with time series data and could learn from long sequences of inputs. Therefore, I decided to use LSTM model for my project.

To sum up, this project successfully showed the usage of machine learning techniques for stock price analysis and prediction. From the obtained results, one may conclude that despite being useful for providing insights, the models cannot fully comprehend the nature of financial markets' unpredictability.

It was shown that LSTM technique outperforms linear regression, which is why the former should be used for time series analysis. Finally, the limitations of this research point at the necessity to approach the usage of such tools carefully.

Apple stock prices were also predicted to have an upward trend future by the prediction dashboard because the LSTM model accurately reflected the real stock prices and predicted an upward movement in the future even when there were some market volatility issues.

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APPENDIX

PowerBI dashboard link: https://londonmet-my.sharepoint.com/:u:/r/personal/bas0600_my_londonmet_ac_uk/Documents/CS7P01-MSc-Project-powerbi-dashboard-24048478.pbix?csf=1&web=1&e=tBb5GJ

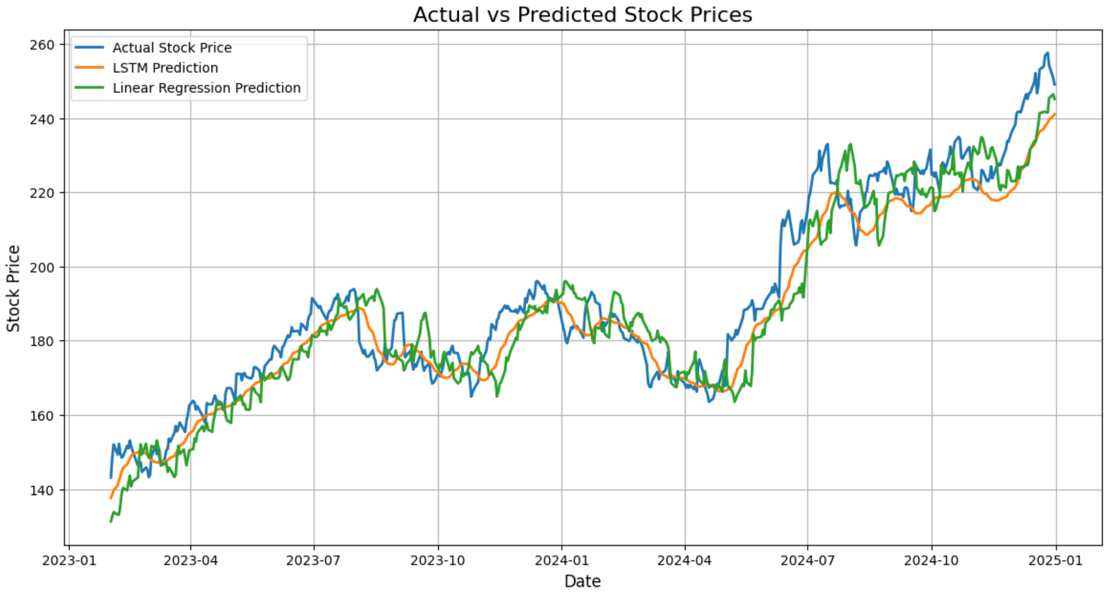

Figure 26: Actual vs Predicted Stock Price of Apple

```
... Statistical Summary:
              index                           Date        Close         High  \
count  2467.00000                           2467  2467.000000  2467.000000
mean   1282.00000  2020-02-05 07:46:22.813133312    95.106605    96.029662
min      49.00000            2015-03-16 00:00:00    20.584810    20.887868
25%     665.50000            2017-08-23 12:00:00    36.142017    36.323360
50%    1282.00000            2020-02-06 00:00:00    67.593903    68.519553
75%    1898.50000            2022-07-19 12:00:00   151.102249   152.592951
max    2515.00000            2024-12-31 00:00:00   257.612701   258.686851
std     712.30588                            NaN    65.328540    65.920667

               Low         Open        Volume         MA20         MA50  \
count  2467.000000  2467.000000  2.467000e+03  2467.000000  2467.000000
mean     94.079422    95.017160  1.145220e+08    94.248693    92.978465
min      20.386574    20.507344  2.323470e+07    21.386588    21.805716
25%      35.811303    36.045637  7.045165e+07    36.159532    35.066364
50%      67.147956    67.622832  9.850720e+07    66.054605    66.234212
75%     148.949537   150.826186  1.387928e+08   148.922099   149.372409
max     256.230269   256.787224  6.488252e+08   248.452274   236.241994
std      64.643514    65.249604  6.569559e+07    64.644110    63.752737

           Returns
count  2467.000000
mean      0.001055
min      -0.128647
25%      -0.007326
50%       0.000941
75%       0.010129
max       0.119809
std       0.017926
```

Figure 27: Cleaned Dataset Head created by python

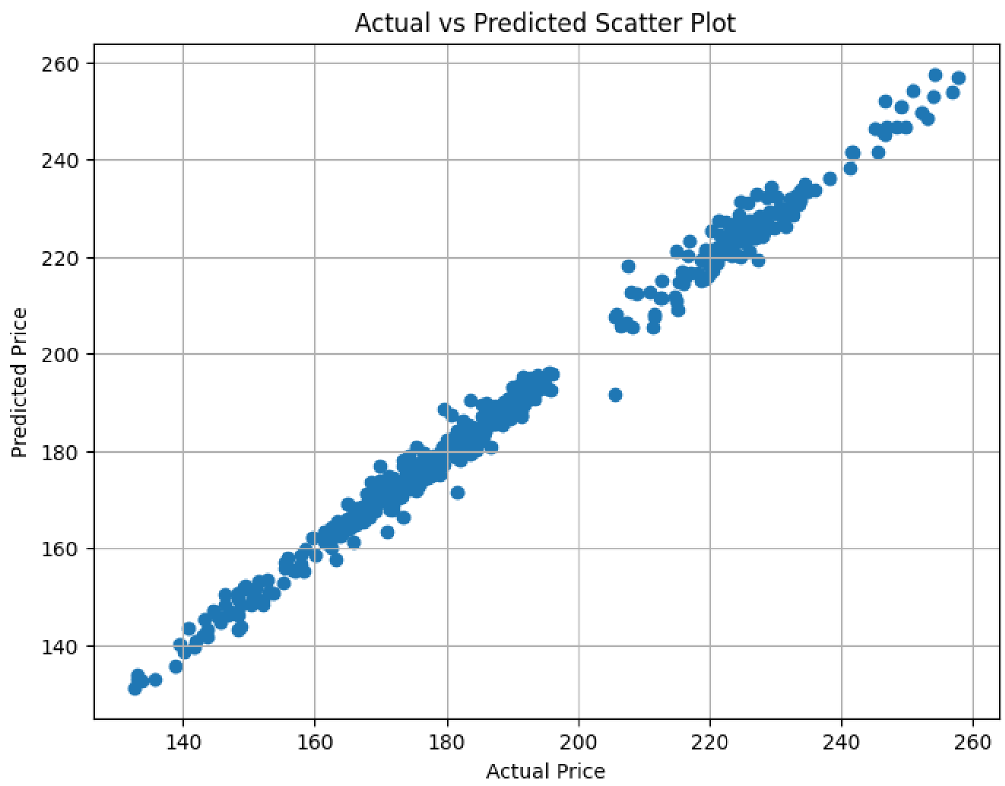

Figure 28: Linear Regression actual vs predicted price scatter plot

```
Missing Values After Cleaning:
index      0
Date       0
Close      0
High       0
Low        0
Open       0
Volume     0
MA20       0
MA50       0
Returns    0
dtype: int64
```

Figure 29: Missing values after data cleaning

```python
# ---------------------
# CREATE predictions.csv
# ---------------------

# Align dates for LSTM
lstm_dates = data['Date'].iloc[-len(lstm_pred):].reset_index(drop=True)

predictions_df = pd.DataFrame({
    'Date': lstm_dates,
    'Actual Price': y_test_actual.flatten(),
    'Predicted Price (LSTM)': lstm_pred.flatten()
})

# Add Linear Regression predictions (trim if needed)
predictions_df['Predicted Price (Linear Regression)'] = lr_pred[:len(predictions_df)]

# Save file
predictions_df.to_csv("predictions.csv", index=False)

print("predictions.csv created!")
```

```
predictions.csv created!
```

Figure 30: Predictions dataset creation

```python
print("\nFirst 5 Rows of predictions_df:")
print(predictions_df.head())
```

```
First 5 Rows of predictions_df:
        Date  Actual Price  Predicted Price (LSTM)  \
0 2023-02-01    143.134674              137.645721
1 2023-02-02    148.439621              138.602432
2 2023-02-03    152.061523              139.688141
3 2023-02-06    149.335236              140.968704
4 2023-02-07    152.209152              142.239166

   Predicted Price (Linear Regression)
0                           131.328707
1                           132.656770
2                           133.817630
3                           133.099463
4                           133.158517
```

Figure 31: Brief of predictions table created

```python
#======================================================
# APPLE STOCK PRICE ANALYSIS AND PREDICTION MSC-PROJECT
#======================================================

# ----------------
# Import Libraries
# ----------------

import pandas as pd
import numpy as np
import yfinance as yf
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
```

Figure 32: Python libraries integration

Date	Open	High	Low	Close	Volume	MA20	MA50	Returns
16 March 2015	27.5457735734784	27.7836971282959	27.3211926838062	27.7836971282959	143497200	28.3976318359375	26.56214427948	0.0110040795771216
17 March 2015	27.9949435323127	28.3106922858776	27.9393538913417	28.2484321594238	204092400	28.3888485908508	26.6428151321411	0.0167268966754834
18 March 2015	28.2395369895278	28.7198322912101	28.099451708101	28.5664043426514	261083600	28.3860692024231	26.7434886550903	0.0112562772132987
19 March 2015	28.6286638297287	28.7398431067373	28.3284801210975	28.350715637207	183238000	28.3755072593689	26.8398043060303	-0.00755043241904618
20 March 2015	28.5174800773726	28.5508324980722	27.8303930675874	27.9949378967285	274780400	28.3354825019836	26.9224040603638	-0.0125491626042624
23 March 2015	28.2662148206877	28.4285355853744	28.1327983555425	28.286226272583	150838800	28.2711096763611	26.9924907684326	0.010405037401013
24 March 2015	28.2906804395773	28.4707886345128	28.1416989145174	28.1706066131592	131369200	28.2101836204529	27.059733505249	-0.00408748973120865
25 March 2015	28.137254753092	28.199514883696	27.4346008300781	27.4346008300781	206620800	28.1500356674194	27.1244819641113	-0.0261267282308805
26 March 2015	27.2967305228432	27.7681294672599	27.2611523477759	27.6258201599121	190291600	28.081326675415	27.1887580490112	0.00697000590671393
27 March 2015	27.6992005079334	27.7281065045355	27.3300862503658	27.4056873321533	158184800	28.0234021186829	27.2504920196533	-0.00796837257625471
30 March 2015	27.5835774155787	28.1061196424448	27.5724588100548	28.0994491577148	188398800	27.9931614875793	27.3393017196655	0.0253145201998848
31 March 2015	28.0371879544733	28.1261317112254	27.6525086239778	27.6680736541748	168362400	27.9383501052856	27.4231604385376	-0.0153517423462233
01 April 2015	27.7548018205594	27.8215100825965	27.372344737332	27.6280574798584	162485600	27.8906548500061	27.4941258621216	-0.00144629419512798
02 April 2015	27.8014921437962	27.9193419161924	27.6147117565556	27.8659763336182	128880400	27.8785363197327	27.5661729812622	0.00861149409194684
06 April 2015	27.6769679354413	28.3529381052622	27.6458378753443	28.3173599243164	148776000	27.8868745803833	27.6346231842041	0.0161983770205705
07 April 2015	28.3818415144798	28.4885726546536	28.0127272513925	28.0193977355957	140049200	27.8743111610413	27.6945449066162	-0.010522244641346
08 April 2015	27.9838256501008	28.1061235419882	27.7881507195423	27.9282360076904	149316800	27.8864298820496	27.7521117782593	-0.00325352203375384
09 April 2015	27.9838206698098	28.1461431432025	27.7192151726219	28.141695022583	129936000	27.934459400177	27.8314894104004	0.00764312557491276
10 April 2015	28.0060594768211	28.2862317326624	27.8526332266256	28.2617721557617	160752000	27.9639218330383	27.8859375	0.00426687635845502
13 April 2015	28.544168668626	28.5886430971563	28.1528188051854	28.206184387207	145460400	28.0001665115356	27.9233711242676	-0.0019668889922515
14 April 2015	28.2395354973429	28.3040196796129	27.9971654951709	28.0838851928711	102098400	28.0151759147644	27.9660664367676	-0.00433589998055206
15 April 2015	28.1083545390746	28.2684513236966	28.0194107499768	28.1906261444092	115881600	28.0122856140137	28.0043850708008	0.00380079005468881
16 April 2015	28.0794369928347	28.2617709331858	28.0416364470639	28.0549774169922	113476000	27.9867142677307	28.0399021148682	-0.0048118380458142
17 April 2015	27.9171186410108	28.0483093806547	27.6747469219012	27.7392311096191	207828000	27.9561400413513	28.0650731277466	-0.0112545557488778
20 April 2015	27.9215657857156	28.4885791257796	27.8326220177268	28.372953414917	188217200	27.9750408172607	28.0991384124756	0.0228457055205866
21 April 2015	28.4841238998735	28.5063577131002	28.166149495148	28.2195167541504	129740400	27.9717053413391	28.1346266174316	-0.00540784945870076
22 April 2015	28.2373180566734	28.6553516127286	28.0883382022584	28.5997619628906	150618000	27.9931631088257	28.1742065811157	0.0134745471388809

Figure 33: stock_data dataset in powerbi

Date	Actual Price	Predicted Price (LSTM)	Predicted Price (Linear Regression)	Error_LSTM
18 January 2023	133.075973510742	127.120155	127.586468515657	5.95581851074219
19 January 2023	133.135025024414	127.71713	127.901218667973	5.41789502441407
20 January 2023	135.693984985352	128.29974	123.120388349015	7.39424498535155
23 January 2023	138.882843017578	128.90904	124.389401240237	9.97380301757812
24 January 2023	140.280426025391	129.66289	123.071203871778	10.6175360253906
25 January 2023	139.621032714844	130.593	127.596305411104	9.02803271484379
26 January 2023	141.687881469727	131.59685	128.117645618818	10.0910314697266
27 January 2023	143.62678527832	132.61516	128.68820080577	11.0116252783203
30 January 2023	140.743057250977	133.70174	131.403260204265	7.04131725097659
31 January 2023	142.012664794922	134.75165	131.324565040686	7.26101479492186
01 February 2023	143.134674072266	135.65205	132.652591679092	7.48262407226565
02 February 2023	148.439605712891	136.48764	133.813375843891	11.9519657128907
03 February 2023	152.0615234375	137.40189	133.095251974226	14.6596334375
06 February 2023	149.335250854492	138.57155	133.154273346911	10.7637008544922
07 February 2023	152.209167480469	139.86288	135.711927167247	12.3462874804688
08 February 2023	149.522247314453	141.12617	138.89915754722	8.39607731445315
09 February 2023	148.488815307617	142.33463	140.296027202757	6.15418530761718
10 February 2023	148.853546142578	143.31815	139.636970458779	5.53539614257812
13 February 2023	151.652984619141	144.09221	141.702764255743	7.56077461914066
14 February 2023	151.012283325195	144.77934	143.64067841189	6.23294332519532
15 February 2023	153.111846923828	145.4489	140.758422292789	7.66294692382812
16 February 2023	151.514984130859	146.09105	142.027381805508	5.42393413085941
17 February 2023	150.371551513672	146.71114	143.148818388517	3.6604115136719
21 February 2023	146.359680175781	147.19472	148.451042293692	0.835039824218711
22 February 2023	146.78352355957	147.43633	152.071111324352	0.652806440429686
23 February 2023	147.266510009766	147.39949	149.34623028141	0.132979990234361
24 February 2023	144.614959716797	147.23494	152.218680007064	2.61998028320312

Figure 34: Predictions dataset in powerbi

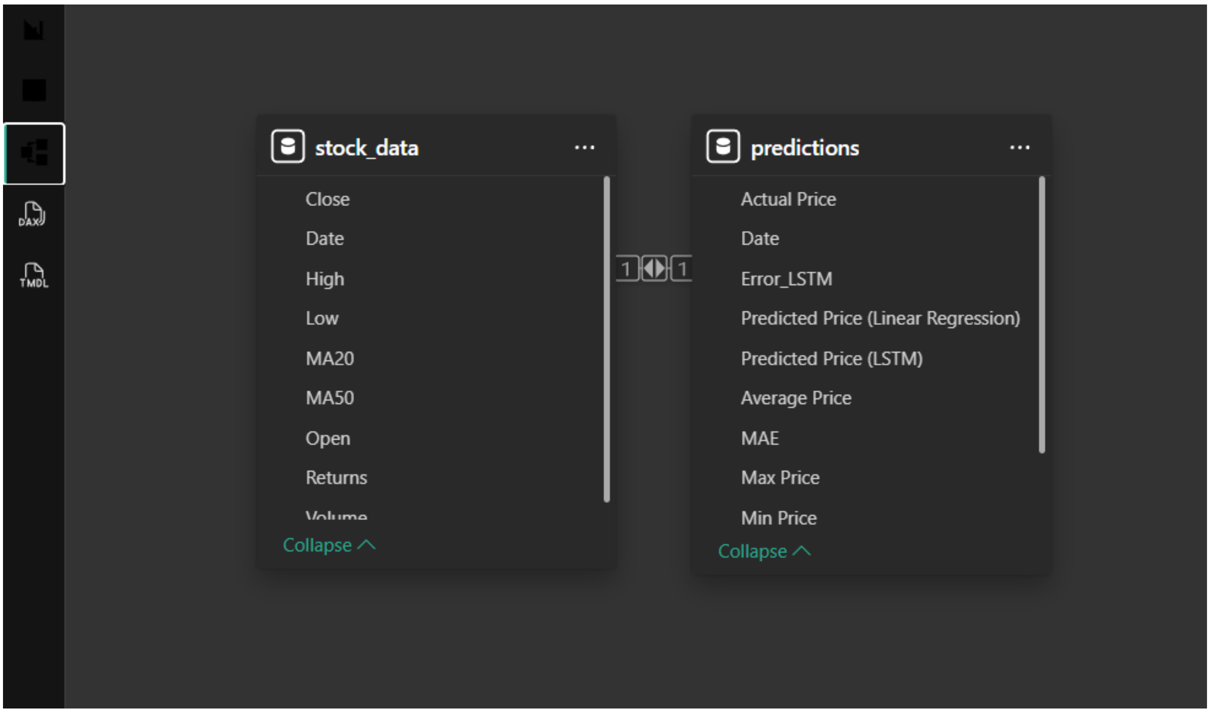

Figure 35: Model view of both the tables in powerbi

```python
# EXPORT CSV
stock_data.to_csv("stock_data.csv", index=False)

print("\nstock_data.csv created successfully!")

# PREVIEW CLEANED DATA HEAD
print("\nCleaned Dataset Preview:")
print(stock_data.head())
```

Figure 36: stock_data dataset generation after cleaning